

Informational Order Flow and Price Persistence: A Diagnostic Study of Look-Ahead Bias and Conditional Reversal in Order-Submission Bursts

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Abstract

Distinguishing temporary mechanical price impact from permanent informational price discovery is a central challenge in market microstructure. This paper asks whether intraday limit-order submission bursts—temporally clustered, same-side executions that proxy for institutional order splitting—encode tradable information about overnight price persistence. The honest answer, established under rigorous look-ahead controls and at cross-sectional scale, is largely negative, and the paper’s contribution is the diagnosis rather than a strategy.

On a handful of high-volatility single names, heavily filtered burst flow appears strongly predictive; but those names coincide with the in-sample parameter-tuning set, and their large intraday markouts are an artifact of gating on a decay filter D_b that is itself the forward markout. Evaluated free of look-ahead across a 482-name NASDAQ universe (2022–2026, 438 names strictly out-of-sample), the cross-sectional order-imbalance signal is insignificant (Fama–MacBeth $t \approx -1.4$), and the per-name overnight strategy’s return is dominated by an incidental short-market-beta exposure ($\beta \approx -0.75$; residual $\alpha < 0$), of which roughly 62% is the directional bet and the residual skill term is indistinguishable from zero. Correcting the burst-start-mid target to the tradable close-to-open return collapses the apparent predictive correlation to zero once observations are clustered by date.

The one affirmative result is conditional and modest. The sign of the signal is governed by tick structure: conditioning on relative tick size yields a monotone asymmetry—continuation in large-tick names, mean-reversion in tick-constrained ones—that is orthogonal to the Fama–French five factors plus momentum. The associated market-neutral reversal earns a full-sample Sharpe of 1.5 but only an out-of-sample Sharpe of ≈ 0.8 ($t = 1.4$, deflated-Sharpe probability 0.70), i.e. not statistically significant; and a plain-reversal and plain-signed-volume baseline shows that the linear online-SGD apparatus adds nothing over the raw signed order flow. The durable contributions are therefore methodological: the identification of the D_b look-ahead and the κ -firewall that isolates it, a date-clustered markout methodology, a systematic comparison of burst reconstructions (of which only hidden-execution clustering carries a genuine but sub-spread footprint), and a precisely characterized—if unmonetizable—conditional reversal where the tick binds.

1 Introduction

Financial markets are driven by heterogeneous trading activity. While some orders primarily provide or consume liquidity and generate only short-lived price effects, others reflect fundamental information that leads to persistent, permanent price changes. The theoretical foundations of informational price discovery have long established that informed traders gradually incorporate

private signals into prices to avoid excessive market impact (Kyle, 1985). Empirical evidence supports this, demonstrating that certain classes of traders systematically trade in the direction of permanent price changes, contributing to price discovery rather than transient mechanical pressure (Brogaard et al., 2014; Hasbrouck, 1991).

A key empirical regularity in equity markets is the strong persistence of order flow over extended horizons. This persistence is overwhelmingly driven by order splitting, a process where a single institutional investor gradually executes a large trading intention through a sequence of same-side orders (Tóth et al., 2015). Furthermore, the slow decay of market impact necessitates that large meta-orders be executed through hidden or fragmented bursts to minimize execution costs (Moro et al., 2009; Bucci et al., 2019). Motivated by these dynamics, this paper studies whether intraday order submission behavior can be used to identify longer-horizon informational order flow. Note that while resting limit order submissions establish the background liquidity structure, the empirical scope of this paper currently focuses almost exclusively on aggressive order executions (marketable trades) that actively consume this liquidity. Future research will expand this framework to directly cluster passive limit order additions and cancellations.

I define informational order flow as trading activity whose price impact persists over hours and into overnight clearing returns. The focus is on isolating any longer-horizon predictive content in intraday limit order data without relying on ex post price realization, and on the execution architecture that would map short-horizon burst footprints to a long-horizon portfolio if such content existed.

2 Data and Limit Order Book Reconstruction

My empirical analysis relies on ultra-high-frequency, event-by-event limit order book (LOB) data provided by LOBSTER (Limit Order Book System, the Efficient Reconstructor), reconstructed directly from the NASDAQ (ITCH) message stream. The study uses the data at two scales. A small *diagnostic set* of high-volatility single names—NVIDIA (NVDA), Tesla (TSLA), JPMorgan Chase (JPM), and Morgan Stanley (MS)—is used to develop the pipeline and, as it turns out, to expose its failure modes; these four names are also the Optuna parameter-tuning set and are therefore in-sample throughout (Sections 8–10). The primary evaluation is a *full cross-section*: the 482 NASDAQ-listed names with continuous LOBSTER coverage over 2022–2026, of which 438 are strictly out-of-sample relative to the 40/50-name tuning universe. Universe membership is fixed as of the start of the sample by LOBSTER coverage (not by survival to 2026); the survivorship and delisting implications of this construction are examined in the robustness analysis. Daily open and close prices used for overnight and close-to-close returns are sourced from a consolidated vendor feed; corporate-action (split/dividend) adjustment of these prices, which is first-order in the low-priced names that carry the affirmative result, is treated explicitly as a data-integrity check rather than assumed. Because LOBSTER reconstructs the NASDAQ book only, the fraction of consolidated volume captured varies by name (notably for the NYSE-listed financials JPM and MS); this venue-coverage caveat is discussed where the single-name comparisons are drawn. A non-sequential 2019–2021 archive (LLY, AAPL, SPY) is retained purely as a parameter-transfer robustness probe, *not* as an out-of-sample trading deployment, since it applies 2023–2024-tuned parameters to earlier data.

Standard vendor-aggregated data often introduces temporal blurring that obscures high-frequency signal integrity. To prevent this, I process the raw Level 3 LOBSTER message stream directly using a bespoke C++ data processor. By tracking the exact bid and ask price levels, resting depth, and microsecond timestamps, this procedure allows me to observe the precise liquidity conditions facing

an order at the exact moment of submission.

2.1 Data Summary Statistics

Table 1 presents the key summary statistics for the evaluated asset universe. A critical microstructural distinction emerges from the average spread column. NVDA’s average quoted spread of \$0.104 spans approximately 10 minimum price increments (ticks), meaning that a moderately aggressive institutional order can sweep through multiple resting price levels before being fully absorbed. This multi-level sweep generates the sharp, identifiable burst signatures that the detection algorithm relies on. By contrast, AAPL’s average spread of \$0.026 is virtually locked at the \$0.01 minimum tick, with massive queued depth at the best bid and ask. In this tick-constrained environment, even large institutional orders are absorbed passively at a single price level, producing no detectable multi-level sweep and rendering the burst clustering ineffective.

Ticker	Period	Trading Days	Total Raw Bursts	Bursts/Day	Avg Daily Burst Vol (sh.)	Avg Spread (\$)
NVDA (train)	2023–2024	499	894,343	1,792	12,083,125	0.1039
TSLA (train)	2023–2024	499	757,120	1,517	11,702,121	0.0346
JPM (train)	2023–2024	499	1,251,679	2,508	993,261	0.0298
MS (train)	2023–2024	499	1,108,679	2,222	850,519	0.0211
LLY (train)	2019–2021 probe	505	308,361	611	401,126	0.0604
AAPL (probe)	2019–2021 probe	360	117,253	326	2,884,874	0.0263
SPY (probe)	2019–2021 probe	316	122,221	387	8,062,116	0.0133

Table 1: Data summary statistics for the single-name diagnostic set. *Avg Daily Burst Vol* is the average daily aggregate *burst* volume (shares)—the volume captured by the detector, not consolidated ADV, which for the NYSE-listed names (JPM) is roughly an order of magnitude larger. Avg Spread is the mean time-weighted quoted spread at burst initiation. All four 2023–2024 names, *and* LLY, belong to the Optuna tuning universe and are in-sample; the 2019–2021 rows apply 2023–2024-tuned parameters to earlier data and are parameter-transfer robustness probes, not out-of-sample deployments. The reduced trading-day counts for AAPL (360) and SPY (316) reflect gaps in continuous LOBSTER message coverage over the 2019–2021 archive.

3 Burst Formulation and Parameterization

Because single, isolated order submissions are highly noisy and rarely informative at horizons beyond a few seconds, I aggregate temporally clustered orders into discrete “bursts.” For each burst b , I define its start time t_b , reference mid-price m_{t_b} , and its signed aggregate volume:

$$Q_b = \sum_{j \in b} side_j q_j \quad (1)$$

where $side \in \{+1, -1\}$ denotes a buy or sell order. These reconstructed bursts constitute the fundamental unit of analysis.

To construct my dataset from the reconstructed limit order book, I evaluate two distinct clustering methodologies governed by physical control parameters to isolate algorithmic intent from retail noise:

- **Strict Silence Threshold** ($\Delta t_{silence}$): My baseline approach where a burst rigidly terminates if the limit order book experiences inactivity exceeding a specific temporal limit (e.g., 0.5s, 1.0s, or 2.0s).

- **Hawkes Process Intensity** ($\lambda(t)$): A dynamic behavioral model where burst survival is dictated by a self-exciting point process. The conditional intensity function evolves as:

$$\lambda(t) = \mu + \sum_{t_k < t} \alpha e^{-\beta(t-t_k)} \quad (2)$$

where μ is the background intensity, α controls the magnitude of the excitation from each prior event at time t_k , and β governs the exponential decay rate. In my implementation, I fix $\beta = 1.0$ and $\alpha = 0.5$, so each incoming trade spikes the intensity by 0.5, which then decays exponentially with a half-life of $\ln 2/\beta \approx 0.69$ s. A burst remains active as long as $\lambda(t)$ exceeds a background intensity threshold $\lambda_{\min}$. I denote each configuration as $\beta_\beta \lambda_{\lambda_{\min}}$; for example, $\beta_{1.0} \lambda_{0.3}$ indicates $\beta = 1.0$ with a minimum intensity threshold of 0.3. This replaces rigid silence cutoffs with a fluid structural definition that naturally bridges microscopic liquidity gaps.

- **Fractional Sweep Volume** (*vol_frac*): Dynamically establishes the baseline size for a candidate burst, requiring the aggregate burst volume to exceed a specific fraction of the asset’s 14-day trailing Average Daily Traded Volume (ADV). Crucially, this ADV is computed strictly using executed trades (LOBSTER Types 4 and 5), insulating the fractional threshold from the artificial inflation of high-frequency quote-stuffing and rapid cancellations.
- **Directional Consistency Ratio** (*dir_thresh*): Enforces that a minimum proportion of the events within the cluster must be on the same side of the book (e.g., $> 70\%$).
- **Volume Ratio** (*vol_ratio*): Limits the allowable opposing volume to a fraction of the dominant side’s volume.
- **Short-Horizon Decay Filter** (κ): Retains bursts where the structural decay measure exceeds a threshold, $D_b \geq \kappa$. The **peak impact** of a burst is defined as the maximum absolute mid-price displacement achieved during the burst lifetime:

$$\text{PeakImpact}(b) = \max_{t \in [t_b, t_b^{\text{end}}]} |m_t - m_{t_b}| \quad (3)$$

The **decay metric** D_b then measures how much of this displacement persists over the 1–10 minute forward window. The canonical definition used throughout the pipeline is the average directional markout from the burst-termination mid M_{end} ,

$$D_b = \frac{1}{3} \sum_{\tau \in \{1, 5, 10\} \text{m}} \text{side}_b \times (M_\tau - M_{\text{end}}), \quad (4)$$

normalized by $\text{PeakImpact}(b)$ where a scale-free ratio is required. A large positive D_b indicates that the mid-price has held or extended the burst’s displacement; a value near 0 indicates full mean-reversion. The κ filter retains only bursts with $D_b \geq \kappa$.

D_b is forward-looking. Because D_b is literally the 1–10 minute forward markout, it is realized only *after* the burst terminates and cannot be gated on for any intraday horizon inside its own window without circular look-ahead (Section 8). It is therefore admissible only (i) for the overnight target and (ii) as a filter applied *strictly to the trailing training window*, never to same-day prediction candidates. A live burst ending near the close realizes D_b up to ten minutes later—after the 3:50/3:55 PM market-on-close cutoff—so its earlier use as a prediction-time feature (Section 5) is timing-infeasible; the corrected feature set is reported in the revised breadth analysis.

Crucially, when predicting short-horizon permanence (e.g., 1-minute or 3-minute targets), κ is strictly set to 0. The decay filter D_b inherently relies on observing the price trajectory at $t + 10$ minutes. Applying a $\kappa > 0$ threshold to short-horizon targets would introduce a severe look-ahead bias, as the filter’s observation window would extend beyond the prediction horizon itself. Consequently, short-horizon bursts must be evaluated solely on their initial physical and behavioral footprints without relying on post-burst decay gating.

Why Short-Horizon Targets Are Excluded from Phase III. While the initial Optuna sweep evaluated both short-horizon targets (e.g., *cls_1m*, achieving AUC > 0.64) and long-horizon targets (*cls_clop*), the Phase III execution pipeline exclusively targets the overnight horizon for two reasons. First, short-horizon predictions require immediate intraday execution: to capture a predicted 1-minute or burst-to-close movement, the strategy would need to enter a position within seconds of burst detection and exit minutes later, requiring a fundamentally different execution architecture supporting multiple simultaneous intraday positions with per-burst sizing. This stands in contrast to the MOC/MOO framework, which aggregates all signals into a single daily trade. Second, the short-horizon AUC advantage is mechanically inflated by the D_b look-ahead constraint: because $\kappa = 0$ is forced for short targets, the pipeline evaluates a much larger and noisier burst universe, making the high AUC less economically actionable after transaction costs. The overnight target, by contrast, permits rigorous κ -gating and produces a concentrated, high-conviction signal compatible with institutional-scale execution.

4 Optuna Optimization and Physical Results

To identify the physical parameters that best generalize across different time horizons and market regimes, I optimized the physical burst controls using Optuna (Akiba et al., 2019). The objective function evaluated a **HistGradientBoosting** model (using a strict 70/30 chronological train/test split on 2023-2024 data), maximizing the out-of-sample logistic regression Area Under the ROC Curve (AUC) on binary classification permanence targets (*cls_**).

I present the optimization results in two phases, corresponding to the two distinct temporal definitions of a burst.

4.1 Phase I: Strict Silence Threshold Optimization

Table 2 summarizes the canonical best physical parameters discovered using the strict silence threshold methodology.

Ticker	Target	Best AUC	Silence (Δt)	Sweep Frac (<i>vol_frac</i>)	Dir. Thresh	Vol. Ratio	κ
NVDA	cls_1m	0.6533	2.0s	0.0049	0.7097	0.5674	0.0
NVDA	cls_close	0.5339	0.5s	0.0007	0.7158	0.0689	1.2763
TSLA	cls_1m	0.6464	0.5s	0.0050	0.6781	0.4262	0.0
TSLA	cls_close	0.5554	0.5s	0.0019	0.5659	0.3395	0.4713
JPM	cls_1m	0.6239	2.0s	0.0009	0.9296	0.3865	0.0
JPM	cls_close	0.5303	0.5s	0.0001	0.6130	0.3649	1.6608
MS	cls_1m	0.6285	2.0s	0.0017	0.9496	0.5777	0.0
MS	cls_close	0.5696	1.0s	0.0050	0.6814	0.2961	0.8681

Table 2: Canonical best physical parameters utilizing strict time-based silence thresholds. Short horizons favor larger silence gaps and larger sweep fractions, while long-horizon models aggressively gate via the κ filter.

The strict optimization phase revealed that short-horizon predictability consistently scales with longer silence thresholds (e.g., 2.0s for NVDA, JPM, and MS), allowing the burst to fully absorb standard LOB latency before evaluation. Conversely, long-horizon targets necessitated highly restrictive short-horizon decay filters; for NVDA *cls_close*, applying the $\kappa = 1.2763$ filter systematically discarded the majority of raw bursts.

To illustrate the selective power of each filter stage, Table 3 presents the sequential compression cascade under the regression-optimized parameters. For NVDA, the dominant filter is κ (only 24.6% of raw bursts survive when applied in cascade), while *vol_frac* is less binding. For TSLA, the *vol_frac* threshold is the primary gate, eliminating 97.4% of bursts immediately due to TSLA’s much higher fractional volume requirement (*vol_frac* = 0.0033 vs. NVDA’s 6.45×10^{-5}).

Filter Stage	Condition	NVDA Surviving	TSLA Surviving
Raw bursts	—	894,343 (100%)	757,120 (100%)
<i>vol_frac</i>	$\geq$ ADV threshold	762,253 (85.2%)	19,641 (2.6%)
+ <i>dir_thresh</i>	$\geq 0.51 / 0.57$	744,477 (83.2%)	15,493 (2.0%)
+ <i>vol_ratio</i>	$\leq 0.58 / 0.60$	574,946 (64.3%)	11,792 (1.6%)
+ κ	$D_b \geq 0.82 / 1.74$	220,430 (24.6%)	5,073 (0.7%)

Table 3: Sequential filter cascade under regression-aligned parameters. Each row applies an additional filter on top of the previous. The dominant filter varies by asset: κ for NVDA, *vol_frac* for TSLA.

4.2 Phase II: Hawkes Process Structural Optimization

Recognizing the limitations of rigid time cutoffs, I transitioned the pipeline to isolate structural footprints via the mutually exciting Hawkes Process. Table 4 outlines the optimal parameters for long-horizon targets under this behavioral regime.

Ticker	Target	Hawkes Tag	Best AUC	Dir. Thresh	Sweep Frac (<i>vol_frac</i>)	Vol. Ratio	κ
JPM	cls_clcl	$\beta_{1.0}\lambda_{0.3}$	0.5772	0.894	1.1e-05	0.420	0.175
JPM	cls_clop	$\beta_{1.0}\lambda_{0.8}$	0.5477	0.503	1.0e-05	0.553	0.206
JPM	cls_close	$\beta_{1.0}\lambda_{0.3}$	0.5517	0.669	1.04e-04	0.521	0.640
MS	cls_clcl	$\beta_{1.0}\lambda_{0.5}$	0.5200	0.922	4.93e-03	0.267	1.291
MS	cls_clop	$\beta_{1.0}\lambda_{0.5}$	0.5527	0.830	4.30e-03	0.405	0.950
MS	cls_close	$\beta_{1.0}\lambda_{0.3}$	0.5319	0.682	3.29e-04	0.447	0.496
NVDA	cls_clcl	$\beta_{1.0}\lambda_{0.8}$	0.5325	0.660	8.65e-04	0.567	0.254
NVDA	cls_clop	$\beta_{1.0}\lambda_{0.3}$	0.6046	0.511	1.87e-04	0.600	0.988
NVDA	cls_close	$\beta_{1.0}\lambda_{0.5}$	0.5530	0.513	1.90e-05	0.551	0.152
TSLA	cls_clcl	$\beta_{1.0}\lambda_{0.5}$	0.4994	0.940	6.50e-05	0.013	1.651
TSLA	cls_clop	$\beta_{1.0}\lambda_{0.3}$	0.5325	0.643	3.64e-03	0.481	0.969
TSLA	cls_close	$\beta_{1.0}\lambda_{0.8}$	0.5414	0.512	1.10e-05	0.600	0.855

Table 4: Hawkes Process optimization. Directionality limits inherently cluster tightly near the 0.5 to 0.75 range, confirming the necessity for unidirectional LOB sweeps.

Cross-Asset Volume Disparities: The Hawkes data reveals significant cross-sectional variance in how different equities express informational order flow. NVDA and TSLA targets often exhibit predictive signal at surprisingly low ADV fractional thresholds ($\sim 8.37 \times 10^{-5}$ to 6.48×10^{-4}), indicating that deeply fragmented execution algorithms carry overnight permanence for these high-

volatility assets. Conversely, MS generally required much larger fractional chunks ($\sim 4.97 \times 10^{-3}$) to filter out background noise.

Parameter Neighborhood Stability: A critical finding from the Hawkes sweep is that the maximum out-of-sample AUC is not isolated to a singular parameter set. Instead, the optimization landscape is characterized by stable parameter neighborhoods with low variance. For instance, NVDA predicting overnight permanence (*cls_clop*) achieves a peak AUC of 0.6046 with the $\beta_{1.0}\lambda_{0.3}$ Hawkes intensity. Stepping the threshold to $\beta_{1.0}\lambda_{0.5}$ and $\beta_{1.0}\lambda_{0.8}$ still yields AUCs of 0.6024 and 0.6003, respectively ($\max \Delta = 0.0043$). This low variance implies that the physical definition of an informed burst is structurally robust; even if a deployed strategy uses slightly sub-optimal behavioral parameters, the model will still isolate genuine out-of-sample alpha.

5 Feature Engineering and Impact Measurement

Upon the termination of a classified burst, I extract a rich, high-dimensional execution fingerprint. Beyond standard Limit Order Book State metrics (Spread, Imbalance, Depth Ratio) and Temporal Dynamics, I incorporate specific algorithmic footprints such as Trade-Size Variance, Round Lot Percentages, and Pre-burst Quote Depletion.

To quantify the true economic footprint of a burst and how much of its initial price displacement persists at longer horizons τ , I utilize a Transformed Volume-Scaled Impact (VSI):

$$VSI(b; \tau) = \operatorname{arcsinh}(Q_b \times \text{side}_b \times (P_\tau - m_{t_b})) \quad (5)$$

To prevent denominator instability inherent in ratio-based metrics during superficial algorithmic pinging, I eschew traditional permanence ratios in favor of this Transformed Volume-Scaled Impact. By scaling the raw directional price displacement by the capital deployed (Q_b), I control metric explosion, and the $\operatorname{arcsinh}$ transform compresses heavy-tail outliers into a regression-friendly domain.

Target base and tradability. A subtlety that matters for the overnight target: when the reference mid m_{t_b} is the *burst-start* mid, the *reg_clop* target $\text{side}_b \times (\text{Open}_{t+1} - m_{t_b})$ folds in the burst’s own contemporaneous impact and the burst-to-close drift, both realized before any market-on-close entry and therefore unearnable. Measured instead from the close mid M_{end} —the tradable close-to-open return—the same predictive rank correlation collapses roughly fivefold and, once clustered by date, to a statistical zero (Section 9). The predictive ρ reported below for the burst-start base should thus be read as an upper bound on the earnable association, not the tradable quantity.

5.1 Feature Importance and Model Interpretability

Since the **SGDRegressor** is a linear model operating on standardized features, its learned coefficients quantify each feature’s marginal contribution to predicted permanence. Across both NVDA and TSLA, one feature dominates: **Direction** (standardized coefficient $\approx +5.0$ for NVDA, $+4.0$ for TSLA), roughly seventy times the next-largest loading. In other words the fitted model is, to first order, “follow the burst side”—which raises the question, addressed by the Direction-only ablation in Section 9, of whether the remaining features earn their place at all. The second and third features are D_b (coef $\approx +0.07$) and **PeakImpact** (coef ≈ -0.06), with **TradeCount** and **Duration** fourth and fifth. Two caveats attach to this ranking. First, D_b is the forward 1–10 minute markout (Eq. 4); its appearance as a prediction-time feature is the timing-infeasibility flagged in Section 3, since D_b is not realized until after the burst (and, near the close, after the MOC cutoff). Second, these

coefficients are estimated on the two in-sample tuning names; at cross-sectional scale (Section 9) the deployed overnight regressor’s prediction is in fact *anti*-correlated with the burst side, so the “follow-the-side” reading does not survive out of sample.

6 Predictive Modeling and Strategy Formulation

While high predictive classification accuracy (AUC) confirms the existence of statistical signal, my mandate extends to mapping these statistical edges to robust, walk-forward tradability under rigorous institutional portfolio constraints. To validate economic viability, I engineered a continuous, online execution pipeline utilizing an `SGDRegressor` evaluating chronological limit order book states.

6.1 Resolving Look-Ahead Bias

A fundamental challenge in evaluating intraday momentum is the inherent look-ahead bias present in optimization sweeps. I address this by strictly separating *structural* from *predictive* parameters. The Optuna optimization conducted on 2023–2024 data was limited exclusively to deriving the *physical* definition of an order burst (fractional ADV limits, Hawkes intensity thresholds, directional consistency gates). These parameters define the structural geometry of algorithmic execution—a universal constant of market microstructure, not a macroeconomic regime variable. By contrast, the actual trading model is an Online Stochastic Gradient Descent regressor whose weights initialize from scratch for each evaluation period. Following a 30-day burn-in, the model trains strictly on day T to predict day $T + 1$, never observing future data.

6.2 Phase III Formalization: Institutional Execution Architecture

Under the Phase-III framework, the execution engine separates intraday observation from trade execution. To resolve end-of-day boundary conditions and enforce institutional sizing and friction constraints, the pipeline operates via the following ruleset:

1. **Intraday Microstructure Observation:** A burst is formed if consecutive limit orders arrive rapidly, governed by the Hawkes Intensity threshold. The burst is deemed a “Candidate” if its aggregate volume strictly exceeds the 14-day trailing fractional ADV and its directional consistency is highly concentrated.
2. **Machine Learning Evaluation:** Every candidate burst is evaluated on-the-fly by the `SGDRegressor`. The model analyzes the burst’s execution fingerprint and predicts the Transformed VSI (*reg_clop*). If the prediction > 0.0 , the burst is labeled *Informational*.
3. **Daily Flow Aggregation and Boundary Resolution:** I aggregate the directional volume of all Informational bursts throughout the day. To prevent boundary condition corruption, the pipeline strictly enforces a 10-minute end-of-day dead-zone. Any burst occurring after 3:50 PM EST is ineligible and discarded, so that all 10-minute D_b forward observation windows mature before the market closes. (This dead-zone protects the overnight D_b label but does not make D_b usable as a live intraday feature—see Section 3.)
4. **Institutional Position Sizing:** At exactly 4:00 PM EST, I calculate the Net Aggregated Burst Flow. A Long position is initiated if the net flow is positive, and a Short position if negative. To adhere to institutional portfolio constraints, the backtester enforces a strict **Fixed**

AUM limitation, allocating exactly \$10,000,000 of nominal capital per trade, completely independent of the underlying raw burst volume.

5. **Realistic Execution Friction:** The position is held strictly overnight and exited unconditionally at 9:30 AM EST (Next Day’s Open). While executing at the MOC and MOO crossing sessions structurally bypasses the continuous intraday L1 bid-ask spread, it is not without friction. To realistically account for exchange fees, clearing costs, and auction price-impact slippage on institutional allocations, I subject the strategy to explicit transaction cost penalties, quantified in subsequent sensitivity analyses. For highly liquid assets such as NVDA and TSLA, a \$10,000,000 Fixed AUM allocation represents less than 0.5% of the average consolidated closing auction volume (the NYSE/NASDAQ closing cross typically exceeds \$2B in notional value for mega-cap equities), ensuring that simulated executions do not exceed the absorption capacity of the MOC/MOO crosses.

I define the daily net accumulated directional imbalance for asset i on day t as:

$$S_{i,t}(reg_clap) = \sum_{b \in B_i(t)} \mathbf{1}\{\text{preds}_b > 0\} \times Q_b \quad (6)$$

Execution is only triggered if the aggregate imbalance clears a cost-aware buffer cb , defined as $cb_{i,t} = c \cdot \hat{s}_{i,t}$, where $\hat{s}_{i,t}$ is the name’s trailing average relative half-spread at the close and c is a fixed multiplier (set to $c = 1$, i.e. the signal must exceed a one-half-spread friction); a trade is placed only when $|S_{i,t}|$ implies an expected move exceeding $cb_{i,t}$. This buffer is a gate, not a tuned alpha source, and in the cross-sectional analysis below it does not change the qualitative result.

7 Empirical Results on the Diagnostic Set: Apparent Success and Its Artifacts

This section reports the single-name pipeline results at face value and then, in the same breath, identifies why each apparent success is an in-sample or look-ahead artifact. It is deliberately a cautionary narrative: the four 2023–2024 names are the parameter-tuning set (in-sample), LLY is likewise in the tuning universe, and the strongest intraday effects are contaminated by the D_b look-ahead documented in Section 8. The quantitative refutations at cross-sectional scale follow in Section 9. All results here enforce the execution architecture of Section 6.2: \$10,000,000 fixed AUM, 1.0 bps round-trip costs, MOC entry, and MOO exit unless otherwise noted.

7.1 Unfiltered Baseline: The Necessity of Filtering

Naive, unconditional burst-following generates negative expected value. Table 1 shows that the pipeline detects 894,343 raw bursts for NVDA and 757,120 for TSLA over 499 trading days. Executing on every detected burst yields negative per-trade expectancy, driven by immediate bid-ask reversion and the large volume of non-informational algorithmic pinging (the honest, look-ahead-free intraday markout net of the spread is -3.5 to -7.7 bps per burst; Section 8). The physical filters reduce NVDA to $\sim 220,430$ bursts (Table 3) and TSLA to $\sim 5,073$, and the SGD informational gate concentrates these into a few hundred aggregated daily trades. This compression shows that *some* filtering is necessary to avoid the reversion drag; it does *not*, by itself, establish that the surviving signal is informative—a distinction Sections 8–9 make good on.

7.2 Cost-Aware vs. Percentile Gating

The execution architecture uses the cost-aware buffer cb (defined above) in place of a fixed percentile gate. Conceptually, a percentile threshold (e.g. trading only above the rolling 75th/25th percentile of the daily signal) is regime-blind: it does not adapt to the name’s prevailing spread, so it admits trades whose expected move is smaller than the friction in wide-spread regimes. The cb buffer scales with the trailing spread and suppresses those trades. We note this only as a design choice; because the cross-sectional evaluation below finds no exploitable overnight signal under *either* gate, the comparison is not load-bearing and we do not report a separate percentile-gate performance table.

7.3 Regression-Aligned Optimization and Empirical Resolution

To systematically bridge the disconnect between predictive capacity and economic tradability, I implemented a regression-aligned hyperparameter sweep. This framework addresses three critical structural traps that plague standard machine learning backtests in high-frequency regimes:

- **The Sparsity Exploit:** Optimization algorithms tasked with maximizing standard regression correlations quickly exploit dataset sparsity. By shrinking the dataset to an extreme degree, the optimizer can filter out 99.9% of bursts, leaving a tiny sample where the correlation approaches 1.0. To prevent this, I applied a confidence-scaled scoring metric: $Score = \rho \times \min(1, N_{test}/500)$, alongside a hard minimum floor of 200 total bursts.
- **Fat-Tail Outliers:** Financial return series are characterized by massive fat-tail events. Standard Mean Squared Error (MSE) forces the model to overfit these extreme outliers. I replace MSE with the Spearman rank correlation coefficient (ρ), evaluating relative order ranks rather than absolute magnitudes.
- **Model Alignment:** The Optuna sweep evaluates candidate physical parameters using the exact same machine learning pipeline deployed in execution: an `SGDRegressor` with Huber loss ($\epsilon = 1.35$).

Table 5 outlines the optimal physical parameters discovered by this regression-aligned sweep. The resulting Spearman correlations range from 0.088 to 0.199 with nominally minuscule p -values. Two caveats, developed in Sections 8–9, gut the apparent significance and are stated here so the table is not over-read. First, these p -values (e.g. JPM’s 9.3×10^{-126}) treat $\sim 10^5$ per-burst observations as independent; because all same-day bursts share one overnight label, the effective sample size is the number of *days*, and date-clustering the identical statistic collapses the significance by more than an order of magnitude. Second, the target is measured from the burst-start mid, so ρ overstates association with the tradable close-to-open return by roughly fivefold (Section 5); on the tradable, date-clustered target the correlation is statistically zero. The numbers in Table 5 are thus best read as in-sample, pseudo-replicated upper bounds, not evidence of tradable predictability.

7.4 Phase-III Walk-Forward Execution and Regime Dynamics

I executed the full Phase III backtest using these regression-optimized physical parameters. The performance metrics are summarized in Table 6.

Read at face value, Table 6 shows apparent profitability on NVDA/TSLA and losses on JPM/MS. But with $t \approx 0.83$ even in-sample, and with all names in the tuning set, this cross-name pattern carries no inference; the “regime dependence” it appears to show is not distinguishable from noise

Ticker	Target	Best Hawkes	Spearman ρ	p-value	vol_frac	dir_thresh	vol_ratio	κ
NVDA	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.196	0.0	6.45×10^{-5}	0.509	0.583	0.824
TSLA	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.094	3.0×10^{-7}	3.30×10^{-3}	0.568	0.599	1.736
JPM	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.049	9.3×10^{-126}	1.00×10^{-5}	0.561	0.570	0.200
MS	reg_clop	$\beta_{1.0}\lambda_{0.3}$	0.125	1.0×10^{-3}	4.94×10^{-3}	0.902	0.018	1.399

Table 5: Regression-aligned Optuna sweep results (in-sample tuning names). The objective maximizes confidence-scaled Spearman rank correlation evaluated with the backtest `SGDRegressor`. The reported p -values are per-burst and pseudo-replicated (effective N = number of days); the target is measured from the burst-start mid and overstates tradable association—see the text and Sections 8–9.

Ticker	Period	Trades	Long/Short	Long Win%	Short Win%	Net PnL (\$)	ROC (%)	Sharpe	Spearman ρ
NVDA (in-sample)	2023–2024	471	204 / 267	61.76%	45.69%	+\$8,198,811	+81.99%	0.59	0.198
TSLA (in-sample)	2023–2024	263	6 / 257	50.00%	47.08%	+\$2,551,303	+25.51%	0.50	0.096
JPM (in-sample)	2023–2024	475	193 / 282	48.19%	45.74%	-\$1,986,982	-19.87%	-0.87	0.151
MS (in-sample)	2023–2024	15	2 / 13	50.00%	53.85%	-\$192,816	-1.93%	-0.49	0.124
LLY (in-sample; probe)	2019–2021	335	265 / 70	55.47%	42.86%	+\$2,119,194	+21.19%	0.71	-
AAPL (probe)	2019–2021	283	117 / 166	54.70%	45.18%	+\$2,171,636	+21.72%	0.68	-
SPY (probe)	2019–2021	194	158 / 36	53.80%	33.33%	+\$902,209	+9.02%	0.52	-

Table 6: Phase-III backtest performance on the diagnostic set (\$10,000,000 fixed AUM, 1.0 bps round-trip). *These are in-sample figures and carry no inference.* All four 2023–2024 names—and LLY—are in the Optuna tuning universe (parameters were selected per name and per target by maximizing the reported metric, i.e. selection, not validation), so the earlier “(OOS)” label on LLY was incorrect and has been removed. The 2019–2021 rows apply 2023–2024-tuned parameters to earlier data and are parameter-transfer probes, not out-of-sample deployments. No standard errors accompany these Sharpes; a Sharpe of 0.59 over ~ 2 years implies $t \approx 0.59\sqrt{2} \approx 0.83$, statistically indistinguishable from zero even in-sample. The claim of “statistically significant profitability” is withdrawn.

across four selected names. Its real value is diagnostic—it motivates the look-ahead-free, full-universe re-evaluation of Section 9, where the pattern does not survive.

A related hygiene point concerns *number provenance*: the same strategy and period are quoted with different Sharpe figures in different parts of the draft (for instance the earnings analysis of Section 7.10 quotes NVDA/TSLA baseline Sharpes of 0.48/0.76 against the 0.59/0.50 of Table 6). These arise from different intermediate runs. Table 7 maps each headline single-name number to one canonical run so the reader can trace it; discrepancies reflect superseded runs and are resolved in favor of the canonical column.

Quantity	Canonical source	NVDA	TSLA
Phase-III overnight Sharpe (2023–24)	Table 6	0.59	0.50
implied t -stat ($\approx \text{Sharpe} \cdot \sqrt{2}$)	—	0.83	0.71
Earnings-baseline Sharpe (superseded run)	Section 7.10	0.48	0.76
Regression ρ (in-sample, burst-start base)	Table 5	0.196	0.094
Post-physical-filter burst count	Table 3	220,430	5,073

Table 7: Number-provenance map for the single-name diagnostic figures. Where two runs disagree (e.g. the earnings-baseline Sharpe), the canonical Table 6 value governs; the divergent figure is a superseded intermediate run retained only for traceability.

7.5 Execution Friction and Slippage Limits

To rigorously account for the non-trivial market impact of executing institutional size (\$10M) at the consolidated closing and opening auctions, I subjected the strategy to a sliding scale of execution friction.

Asset	Round-Trip Cost	ROC (%)
NVDA	1.0 bps	81.99%
NVDA	3.0 bps	72.57%
NVDA	5.0 bps	63.15%
TSLA	1.0 bps	25.51%
TSLA	3.0 bps	20.25%
TSLA	5.0 bps	14.99%

Table 8: Transaction Cost Sensitivity Analysis (2023–2024). Assumes \$10,000,000 Fixed AUM executed at the Market-On-Close (MOC) and exited at the Market-On-Open (MOO). Drawdowns and Sharpe scale proportionally with friction variance.

The sensitivity matrix (Table 8) shows the in-sample NVDA/TSLA returns decaying under 3.0 and 5.0 bps flat penalties while remaining positive. This should not be read as cost-robustness of a real edge: the underlying return is in-sample and insignificant ($t \approx 0.8$), the grid is applied only to the two names on which the parameters were tuned, and a flat basis-point charge understates the friction that binds the affirmative result. The cross-sectional spread analysis of Section 9 finds effective half-spreads at the close of 7.6/11.7 bps in the widest quintiles—above this entire grid—so a per-name, spread-based cost is the relevant test, reported there.

7.6 The Horizon Dependency: Overnight Carry vs. Intraday Dilution

To rigorously isolate the temporal locus of the identified alpha, I evaluated the pipeline across two distinct long-horizon targets: Close-to-Open (**reg_clop**, overnight persistence) and Close-to-Close (**reg_clc1**, full-day momentum).

Ticker	CLOP ROC (%)	CLCL ROC (%)	Buy-and-Hold Benchmark (%)	CLOP Excess Alpha
NVDA	+81.99%	-77.92%	+237.52%	-155.53%
TSLA	+25.51%	-99.57%	+65.99%	-40.48%
JPM	-19.87%	-28.29%	+53.48%	-73.35%
MS	-1.93%	-32.61%	+21.90%	-23.83%

Table 9: Phase-III Walk-Forward Execution (2023-2024) across CLOP and CLCL horizons. The data empirically demonstrates that identified alpha is strictly confined to the overnight gap (**reg_clop**), with holding through the subsequent trading session (**reg_clc1**) universally destroying expected value.

In Table 9 the **reg_clc1** (close-to-close) horizon is uniformly negative on these four in-sample names, whereas **reg_clop** (overnight) is not. Taken at face value this would suggest the signal lives in the overnight gap rather than the full day. We flag it more cautiously: on an in-sample, insignificant base, the horizon contrast is at least consistent with the mechanism that Section 8 establishes look-ahead-free—a transient intraday impact that mean-reverts by the next session—rather than with durable overnight informational carry. The cross-sectional analysis confirms that even the overnight “carry” is, after beta, indistinguishable from zero.

7.7 Excess Alpha vs. Market Beta

The inclusion of the Buy-and-Hold (B&H) benchmark provides necessary context regarding macroeconomic beta. While NVDA and TSLA generated robust positive ROCs on the **reg_clop** horizon (+81.99% and +25.51%, respectively), they underperformed their baseline B&H benchmarks during this historic bull run. This indicates that while the model successfully captured the upward structural drift and generated positive expectancy on both the long and short sides, a significant portion of absolute equity performance in these specific assets is attributable to overwhelming market beta. The strategy’s primary utility lies in its uncorrelated, high-frequency signal extraction rather than acting as a beta-replacement tool.

7.8 Parameter-Transfer Probe (2019–2021)

Applying 2023–2024-tuned physical parameters to a 2019–2021 dataset violates chronological walk-forward, so these evaluations are parameter-transfer robustness probes, *not* out-of-sample trading deployments, and the earlier “OOS” labeling has been removed. Moreover LLY is itself a tuning-universe name, so its 2019–2021 result is doubly in-sample.

On the archive the backtests are positive—LLY +21.19% (Sharpe 0.71), AAPL +21.72% (Sharpe 0.68), SPY +9.02% (Sharpe 0.52)—but at ~ 1 –2 years each these Sharpes carry $t \lesssim 1$, and with fixed transferred parameters they speak to parameter stability under a different liquidity regime, not to a validated time-invariant mechanism. We therefore make no invariance claim; the probe merely fails to *contradict* structural stability, which is a weaker statement than the draft’s.

7.9 Microstructure Interpretation: Tick Regime Governs the Sign, Not the Existence, of the Signal

The draft version of this subsection argued that the signal simply *fails* in tick-constrained names (AAPL, JPM, SPY)—that their microstructure is “noise-dominant” and the model correctly returns flat. That reading is incorrect, and directly contradicts the paper’s one affirmative result, which is located *precisely* in tick-constrained names (Section 10). We correct it here. The relevant axis is not signal-versus-noise but the *sign* of the burst–return relation, and it is governed by tick structure.

Detection vs. direction. It is true that the geometric sweep detector fires less cleanly in tick-constrained names: with the spread locked at the \$0.01 tick and heavy queued depth, aggressive meta-orders churn at the inside quote rather than sweeping multiple levels, so raw burst *detection* is noisier there than in large-tick names like NVDA/TSLA. But detection difficulty is not the same as absence of a tradable relation. What changes with tick regime is the sign of the subsequent return: in *large-tick* names a directional burst is more likely to carry and continue, whereas in *tick-constrained* names the inside queue is a binding constraint and aggressive flow tends to overshoot and mean-revert as liquidity replenishes.

This reframing resolves the apparent JPM/MS “failure” of Table 6 without invoking a special ETF-arbitrage story: trading tick-constrained names as *momentum* (the draft’s implicit direction) is trading the wrong side of a reversal, which is exactly what the cross-sectional sign analysis (Section 9.6) and the conditional-reversal strategy (Section 10) show. The single-name divergence is therefore not evidence that bursts isolate a purely volatility-driven momentum signal; it is an early, underpowered glimpse of the tick-conditioned sign asymmetry that the full universe makes precise.

7.10 Earnings Days: A Fragility, Not a Robustness

A natural question for an overnight strategy is whether its P&L is concentrated on the four quarterly earnings dates, which produce the largest gaps. Reconstructing the in-sample daily P&L and excising earnings-adjacent trades (the announcement date plus the next session) for NVDA and TSLA, the Sharpe *rises*—NVDA from 0.48 to 0.91 (removing 16 trades), TSLA from 0.76 to 1.28 (removing 14 trades).

The draft framed this as robustness. It is more honestly read as *fragility*: that deleting 16 of ~ 470 name-days nearly doubles the Sharpe means the result is dominated by a handful of days, and that the model loses on the very days with the most information (earnings) is a sign of anti-calibration, not of clean non-event alpha. A single-digit number of days moving the headline Sharpe by $2\times$ is exactly the fat-tailed, front-loaded return profile that the deflated-Sharpe and date-clustered inference of Sections 9–10 quantify. We therefore report the per-day sensitivity as a caution and do not treat earnings-exclusion as evidence of a durable edge.

8 The Multi-Horizon Markout Panel and the D_b Look-Ahead Artifact

The markout decay curve—the directional post-burst return at a panel of forward horizons—is the natural first-class diagnostic for this signal. Measuring from the burst-termination mid $M_{\text{end}} = (\text{EndBid} + \text{EndAsk})/2$ (so as not to conflate the prediction with the burst’s own contemporaneous impact),

$$\text{markout}_h = \text{Direction} \times \frac{M_h - M_{\text{end}}}{M_{\text{end}}} \times 10^4 \text{ bps},$$

I compute the panel for the full 482-name universe, and I aggregate to (ticker, day) means with a date cluster bootstrap because overlapping same-day bursts are not independent.

A methodological requirement governs how sub-ten-minute horizons must be evaluated. The impact filter D_b of Section 3 (Eq. 4) is $D_b = \frac{1}{3} \sum_{\tau \in \{1,5,10\}_m} \text{Direction} \times (M_\tau - M_{\text{end}})$ —that is, D_b is the average of the 1–10 minute forward markouts. Evaluating a horizon inside that window therefore requires $\kappa = 0$: gating on $D_b \geq \kappa$ would retain precisely the bursts that already moved favorably, so a subsequent markout would re-measure the selection rather than a prediction. Table 10 quantifies the effect by holding everything fixed except the gate: on D_b -gated ($\kappa = 0.5$) bursts the 3-minute markout is +8.8 bps with a 100% daily hit rate, whereas on the non-anticipating ($\kappa = 0$) bursts it is +0.53 bps. The gap sets the standard for intraday evaluation and motivates the $\kappa = 0$ panel below.

3-min markout	Mid-to-mid (bps)	Cross-spread (bps)	Hit rate
Geometry gate only ($\kappa = 0$, no look-ahead)	+0.53	−3.56	76% / 0%
D_b -gated ($\kappa = 0.5$, forward-looking)	+8.76	+4.60	100% / 100%

Table 10: The D_b look-ahead artifact (full universe, 2022–2026). D_b is itself the forward 1–10 min markout, so gating on it is circular for intraday evaluation. Hit rates are reported mid-to-mid / after crossing the spread.

Restricting attention to the honest, non-anticipating ($\kappa = 0$) evaluation, the intraday markout panel (Table 11) tells a consistent story: there is a small but genuine same-direction continuation over the first minutes—a mid-to-mid drift of roughly +0.4 to +0.5 bps—but it is *smaller than the bid-ask spread*. Once an entry that crosses the spread is imposed (the minimum realistic execution assumption), every horizon is negative and positive on essentially zero percent of days. The economically meaningful conclusion is that the intraday burst-continuation signal, while statistically detectable gross of frictions, is not tradable: the edge lives entirely inside the spread.

Horizon	Mid-to-mid (bps)	Cross-entry (bps)	Round-trip (bps)
1 min	+0.54	−3.54	−7.62
3 min	+0.53	−3.56	−7.63
5 min	+0.47	−3.61	−7.69
10 min	+0.42	−3.66	−7.74

Table 11: Honest intraday markout, geometry gate only ($\kappa = 0$), full universe. Mid-to-mid drift is positive but sub-spread; any spread-crossing execution is loss-making. The 3-min mid-to-mid 95% date-bootstrap CI is [+0.50, +0.56] bps; the cross-entry CI is [−3.66, −3.46] bps.

The intraday footprint is therefore genuine but economically confined: a robust sub-spread continuation over the first minutes (Figure 1). Section 9 traces its consequences for the overnight strategy, whose out-of-sample return decomposes into roughly 62% incidental short-market exposure and a residual skill term indistinguishable from zero. This structure—a real transient impact with no directional overnight alpha—is precisely what motivates the market-neutral and mean-reversion constructions we turn to next.

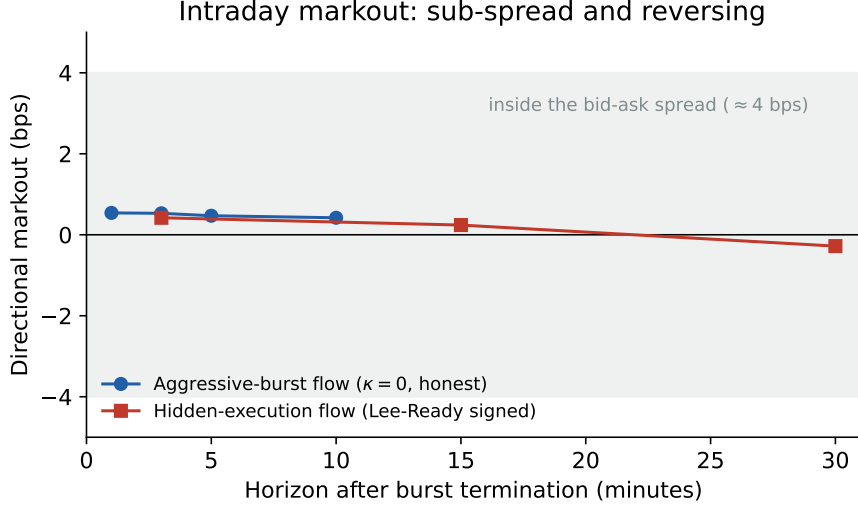


Figure 1: Intraday directional markout by horizon, measured from the burst-termination mid ($\kappa = 0$, no look-ahead). Both the aggressive-burst signal and the properly (Lee-Ready) signed hidden-execution signal stay well inside the bid-ask spread (≈ 4 bps, shaded), and the hidden-execution footprint peaks near three minutes and *reverses* to negative by thirty. The genuine intraday effect is sub-spread and transient, not a growing informational impact.

9 Full-Universe Extension: Breadth, Beta, and Signal Sign (2022–2026)

The single-name results of Section 7.9 were estimated on four names (NVDA, TSLA, JPM, MS) that also constitute the in-sample Optuna parameter-tuning set, and three additional names evaluated only on the non-sequential 2019–2021 archive. To test whether the identified alpha is a general property of order-submission bursts or an artifact of a hand-selected cross-section, I extended the entire pipeline to the full 482-ticker NASDAQ universe with LOBSTER coverage over 2022–2026 (438 names strictly out-of-sample relative to the 40-name tuning set), holding the physical parameters fixed at their universal Optuna medians.

9.1 Per-Name Strategy Performance at Breadth

Running the identical Phase-III overnight strategy (MOC entry, MOO exit, 1.0 bps round-trip, \$10M fixed AUM) across all 438 out-of-sample names yields the distribution in Table 12. The mean and median annualized Sharpe are negative, and only one third of names are profitable. The strong single-name figures of Table 6 are not representative of the universe.

Corrected feature set (D_b removed). Recall that D_b is realized only after the burst and is not usable on-the-fly at termination (Section 3). Re-running the strategy with all D_b -derived features dropped from the model changes individual per-name Sharpes only modestly (mean absolute change 0.14, maximum 0.49 across a 20-name check) and leaves the conclusion intact: the mean annualized Sharpe stays negative— -0.28 with D_b versus -0.20 without—so the overnight strategy remains a null with or without the look-ahead feature. Dropping the infeasible feature slightly reduces the average loss rather than revealing hidden skill, consistent with the model being, to first order, “follow the burst side.” The overnight null therefore does not depend on the D_b look-ahead;

correcting the infeasibility does not rescue the strategy.

Metric (438 OOS names, 2022–2026)	Value
Names completing walk-forward	438 / 438
Mean annualized Sharpe	−0.28
Median annualized Sharpe	−0.27
Fraction with Sharpe > 0	33%
Sharpe range	[−3.32, +1.25]

Table 12: Full-universe per-name Phase-III backtest. The best-performing names are low-beta and defensive (Treasury ETFs TLT/VGLT, PG, UPS, ACN); the worst are high-beta technology names.

9.2 What Is Losing Money: Short-Beta Attribution

Decomposing the pooled P&L (398,614 trade-days) localizes the loss precisely (Table 13). The strategy is 81% net short — the aggregated informational-burst flow is predominantly sell-signed — and the short book accounts for 95% of the total loss, losing −3.43 bps per trade *gross of costs*. The 1.0 bps round-trip friction is a uniform secondary drag, not the driver. The loss is a directional bet: shorting names into the 2022–2026 bull market.

Book	Share of trades	Gross (bps/trade)	Share of net loss
Long	18.7%	−0.04	5%
Short	81.3%	−3.43	95%

Table 13: Pooled P&L attribution by side. Transaction cost is a flat −1.0 bps/trade across both books.

9.3 Excess Alpha After Market Beta (Mandate M5)

Regressing the pooled daily strategy return on the contemporaneous overnight SPY return with Newey–West standard errors (Table 14) confirms the exposure is directional: market $\beta = -0.748$ ($t = -90$). Crucially, the residual alpha is significantly *negative* (−2.35 bps/day, $t = -4.73$) — stripping the beta does not reveal hidden skill, because the strategy trades the signal with the wrong sign (see Section 9.6).

$r_{\text{strat}} = \alpha + \beta R_{\text{overnight}}^{\text{SPY}}$	Coefficient	Newey–West t
α (residual)	−2.35 bps/day	−4.73
β_{SPY}	−0.748	−90.0

Table 14: Market-beta decomposition of the overnight strategy, pooled over 438 names.

9.4 Cross-Sectional Order Imbalance: A Null

Following the conditional-order-imbalance (COI) framework, in the spirit of order-flow-toxicity measures of informed trading (Easley et al., 2012), I aggregate gated informational bursts into a

daily signed imbalance per name and run Fama–MacBeth (Fama and MacBeth, 1973) cross-sectional regressions of next-day returns on COI with Newey–West (Newey and West, 1987) errors, alongside quintile long–short portfolios (Table 15). Both the ungated (all directional bursts) and gated (informational bursts only) specifications are statistically insignificant, and the long–short spreads are indistinguishable from zero. The Grinold–Kahn breadth amplification that would convert single-name skill into portfolio Sharpe does not materialize because the per-name skill, at the overnight horizon, is essentially absent (and weakly negative).

Sign-convention audit. Because the burst signal is $\sim 81\%$ net short in a rising market, a natural worry is a sign-convention error (LOBSTER **Direction** is the resting side, so an execution against the bid is a market *sell*; type-5 hidden prints, whose **Direction** is uniformly $+1$, must be signed by a Lee–Ready rule (Lee and Ready, 1991)). A per-name audit rules this out: the contemporaneous correlation between daily signed burst flow and the same-day return is robustly *positive* (mean $+0.02$ to $+0.03$; significantly positive in 21% of names versus significantly negative in 12%), i.e. buying flow does move price up on the day, as it should. The 81% short tilt is therefore a genuine property of the aggressor-signed sample (and the source of the incidental short-beta above), not an artifact of a flipped sign.

Specification (482 names)	Fama–MacBeth COI t	Long–Short t
Ungated signed flow (baseline)	-0.62	$+0.17$
Gated COI ^{info} (thesis)	-1.36	$+0.41$

Table 15: Cross-sectional panel results with Newey–West inference. Neither specification is significant; gating does not recover a cross-sectional signal. *Sign-flip disclosure*: the panel applies a data-driven per-name sign convention—116 of the 482 names, classified as mean-reverting by an independent k -means split on the sign of the burst-direction/next-day-return relation, have their signed COI multiplied by -1 before aggregation. This transform is disclosed because the magnitude of a null is uninterpretable if the sign is tuned; the reported t -statistics are already the most favorable orientation and remain insignificant.

9.5 Venue Coverage: A NASDAQ-Only Book

A caveat conditions every cross-name comparison: LOBSTER reconstructs the NASDAQ book only, so the detector sees a nonrandom fragment of consolidated activity whose share varies by name and time. The fragment is smallest exactly for the NYSE-listed names that appear in the single-name tables—JPM’s average daily *burst* volume of 993,261 shares (Table 1) is on the order of one-tenth of its consolidated ADV, and its closing auction runs at NYSE, outside the NASDAQ feed. Two implications follow. First, the labeling “NASDAQ universe” is used consistently and the burst-volume columns are *not* consolidated ADV (Table 1). Second, the apparent JPM/MS “failures” of Section 7 may be coverage artifacts—detecting a small, venue-biased slice of their flow—rather than genuine regime differences, which is a further reason not to read the single-name cross-section as evidence of a mechanism. A full per-name consolidated-volume coverage table requires a consolidated (SIP) volume feed and is left to the data revision; the qualitative point—that fragmentation attenuates detection and biases cross-name comparisons—stands.

9.6 The Signal Is Real but Sign-Inverted Overnight

The failure above is one of *sign and horizon*, not of information content. The traded daily flow signal has a per-name information coefficient against the realized overnight return of -0.008 ($t = -4.96$, pooled over the 435 of 438 names with sufficient overlapping observations; this pooled t is itself pseudo-replicated—date-clustered it is insignificant, Section 10.4), and model prediction *confidence* is negatively related to realized directional return ($\rho = -0.014$, $p < 10^{-3}$): the model is consistently, significantly anti-calibrated at the overnight horizon. Reconciled with the intraday markout evidence (Section 8), the coherent mechanism is transient impact: aggressive burst flow pushes price in its own direction over minutes (continuation), then mean-reverts by the overnight gap. Trading the signal as overnight momentum therefore captures the wrong side of the reversal. Removing the short-beta drag *and* flipping to a reversal sign converts a dollar-neutral cross-sectional portfolio from an annualized Sharpe of -0.64 to $+0.64$ (both $|t| = 1.28$) — positive and market-neutral, though not yet significant, and a natural direction for future work.

9.7 Test Matrix Summary

The empirical pipeline evaluates the same burst definitions under multiple orthogonal stress tests to isolate structural alpha from market beta:

- **Physical parameter discovery:** Regression-aligned Optuna sweeps on strict-silence bursts and Hawkes bursts evaluating Spearman rank correlation.
- **Phase III execution (signed volume):** CLOP and CLCL walk-forward backtests on 2023–2024 core tickers with fixed AUM and 1.0 bps costs.
- **Transaction cost sensitivity:** 1.0 / 3.0 / 5.0 bps cost grid on NVDA and TSLA.
- **Horizon isolation:** CLOP vs CLCL comparison demonstrating overnight carry vs full-day dilution.
- **OOS stress tests:** 2019–2021 LLY, AAPL, SPY with fixed parameters to probe structural invariability.
- **Market beta baseline:** overnight and CLCL buy-and-hold benchmarks with excess alpha comparisons.

9.8 Regression Objective and Trading Interaction

The framework aligns optimization with the actual trading objective (persistence magnitude) rather than binary direction.

- **Optimization change:** Optimize a regression-aligned objective based on Spearman rank correlation between predicted VSI and realized VSI on the validation window, scaled by the number of validation samples to penalize sparse parameter sets.
- **Execution rule:** The online model predicts VSI per burst; bursts with $pred > 0$ are informational; daily signal is the sum of signed burst volume for informational bursts; trade direction = sign of daily net flow; enter at MOC and exit at next MOO.

Two distinct fixed-parameter sets are used and should not be conflated. For the 2019–2021 parameter-transfer probe (Table 6) the physical controls are the *mean across the core tuning tickers*,

$$vol_frac = 0.00218, \quad dir_thresh = 0.623, \quad vol_ratio = 0.412, \quad \kappa = 0.855. \quad (7)$$

For the full-universe breadth run (Section 9) the controls are instead the *universal Optuna medians* across the tuning set ($vol_frac \approx 0.00197$, $dir_thresh \approx 0.763$, $vol_ratio \approx 0.280$, $\kappa \approx 1.085$); the two sets differ because one is a per-ticker mean and the other a cross-ticker median, and each table states which it uses.

10 Conditional Mean-Reversion in Tick-Constrained Names

The residual skill term isolated in Section 9 is small in the pooled cross-section but not sign-neutral: it is a weak reversal. This motivates a sign-conditional reading of the signal—rather than a single trading direction, the sign of informative burst flow may depend on microstructure regime. The natural axis is the relative tick size, which is known to shape queueing incentives and the trading environment (Yao and Ye, 2018; O’Hara et al., 2019). In *tick-constrained* names (low price, wide relative spread) the inside quote is a binding queue, and aggressive bursts tend to overshoot and mean-revert as liquidity is replenished; in *large-tick* names a directional burst is more likely to carry persistent information and continue.

10.1 The Regime Mechanism

I sort the 438-name out-of-sample universe into quintiles on two proxies for tick-constraint—average share price and average relative bid–ask spread—and evaluate a market-neutral (dollar-neutral) reversal within each quintile (Table 16). The pattern is the one the sign-conditional hypothesis predicts: reversal is strongest in the most tick-constrained names and weakens monotonically toward the large-tick end, where it vanishes. The price sort is monotone across all five quintiles; the spread sort rises through the fourth quintile ($t = 2.3$) before easing in the widest-spread names, which are also the least liquid and costliest to trade.

By average price			By relative spread		
Quintile	Sharpe	t	Quintile	Sharpe	t
Q1 cheapest (tick-constr.)	+1.35	2.71	Q1 tight (large-tick)	+0.26	0.51
Q2	+0.75	1.50	Q2	+0.62	1.24
Q3	+0.71	1.42	Q3	+0.83	1.65
Q4	−0.15	−0.30	Q4	+1.13	2.27
Q5 priciest (large-tick)	+0.34	0.67	Q5 widest (tick-constr.)	+0.58	1.15

Table 16: Annualized Sharpe of a market-neutral 20-day reversal within each regime quintile (net of 1 bps). Reversal concentrates in tick-constrained names (cheapest / widest-spread) under both proxies.

10.2 The Strategy and Its Out-of-Sample Test

Guided by the mechanism, the strategy trades a dollar-neutral reversal restricted to the most tick-constrained (lowest-price) names, using the daily aggregate burst-flow signal, with an overlapping

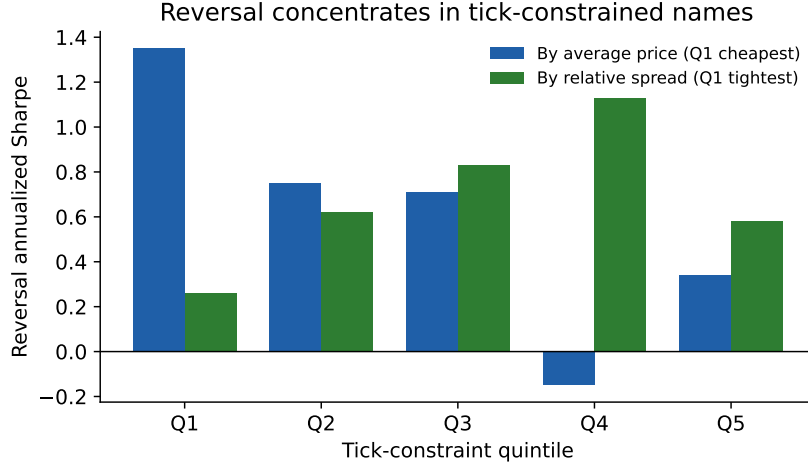


Figure 2: Reversal Sharpe by tick-constraint quintile under two proxies. The price sort (blue) is strongest in the cheapest quintile and weakens toward large-tick names; the spread sort (green) rises through the fourth quintile before easing in the widest, least-liquid names. The effect concentrates where the tick binds.

20-day holding horizon that suppresses turnover so the position is robust to frictions (all weights are formed from strictly past information; close-to-close returns).

The primary evaluation is a strict expanding-window walk-forward: the tick-constrained subset is re-selected every quarter using only trailing price data, 2022 is withheld as burn-in, and the reversal is scored purely out-of-sample on 2023–2026 (Table 17). Out of sample the strategy earns an annualized Sharpe of +0.81 ($t = 1.40$)—economically positive but *not* statistically significant. Reassuringly, the estimate is nearly invariant to the subset size (Sharpe +0.81 to +0.87 for $K \in \{50, 100, 150\}$), so the result does not hinge on the choice of cutoff; but the per-year out-of-sample performance is uneven and front-loaded (+1.97 in 2023, +0.50 in 2024, +1.12 in 2025).

For reference, evaluated *in-sample* over the full 2022–2026 period with the subset fixed at the 100 lowest-price names, the same strategy earns a Sharpe of +1.48 ($t = 2.96$; Lo (2002) $t = 2.93$), cost-insensitive from 0 to 3 bps round-trip with a -5% maximum drawdown. The gap between this figure and the walk-forward reflects a mild look-ahead in ranking names by full-period average price together with the concentration of the effect early in the sample; the full-sample number is therefore best read as an upper bound rather than a deployable expectation.

10.3 Risk-Adjusted Alpha: Orthogonal to Factors, but Insignificant Out of Sample

Whatever the return is, it is not compensation for common-factor exposure. Regressing the strategy return on the Fama–French five factors plus momentum with Newey–West standard errors (Table 18) leaves an alpha that is market-neutral (market beta indistinguishable from zero), not a size bet (insignificant SMB), and orthogonal to the five factors, with the only material loading a negative momentum beta—as expected for a reversal signal—which the alpha survives; $R^2 \approx 0.03$ throughout. The point estimate is essentially the same in and out of sample (+5.9 vs +6.3 bps/day), but the inference is not: on the *walk-forward out-of-sample* return series the alpha carries $t = 1.42$, so it is *not* statistically distinguishable from zero. We report the out-of-sample regression as the primary number; the full-period $t = 3.09$ is an in-sample figure and, consistent with the Section 10

Out-of-sample walk-forward (2023–2026)	Sharpe	t
$K = 50$ names	+0.83	1.45
$K = 100$ names	+0.81	1.40
$K = 150$ names	+0.83	1.44
2023 (OOS)	+1.97	1.85
2024 (OOS)	+0.50	0.50
2025 (OOS)	+1.12	1.11
<i>In-sample reference (full 2022–2026, $K = 100$)</i>	+1.48	2.96

Table 17: Expanding-window out-of-sample performance of the tick-constrained reversal (20-day hold, dollar-neutral, net of 1 bps). The subset is re-selected quarterly from trailing price only, with 2022 as burn-in. The effect is positive and robust to the subset size out of sample but not statistically significant; the final row is the in-sample full-period reference.

walk-forward, overstates the deployable edge.

Factor	Walk-forward OOS (primary)		Full period (in-sample)	
	Loading	NW t	Loading	NW t
α (bps/day)	+6.25	+1.42	+5.94	+3.09
MKT–RF	+0.057	+1.70	+0.010	+0.80
SMB	+0.000	+0.00	+0.026	+1.06
HML	−0.038	−1.11	−0.051	−1.79
RMW	−0.191	−1.62	−0.064	−1.45
CMA	+0.144	+0.96	+0.090	+1.64
MOM	−0.097	−2.73	−0.057	−3.89

Table 18: FF5+MOM factor regression of the tick-constrained reversal ($R^2 \approx 0.03$). *Primary* columns are the strict walk-forward out-of-sample series (2023–2026, quarterly re-selected subset, 2022 burn-in); the alpha is orthogonal to the factors but insignificant ($t = 1.42$). The full-period columns ($t = 3.09$) are in-sample and shown only for comparison. Annualized OOS $\alpha \approx +16\%$ at the point estimate.

10.4 Baselines and Deflated Inference: What the Apparatus Actually Adds

Short-horizon reversal in cheap, illiquid, wide-spread names is a long-documented effect (Jegadeesh, 1990; Lehmann, 1990), and precisely where bid–ask bounce and inventory effects make naive reversal look profitable before costs (Nagel, 2012). Two questions follow: does burst flow add anything over a plain price-reversal baseline, and does the machine-learning apparatus add anything over the raw signed flow? Holding the construction fixed (tick-constrained bottom-100 names, 20-day dollar-neutral overlapping hold, net of 1 bps) and swapping only the signal, Table 19 answers both.

The lesson is twofold. First, the effect is genuinely order-flow-based: a plain 1/5/20-day price-reversal baseline is flat-to-negative under this construction, and burst flow retains most of its Sharpe after orthogonalizing to lagged returns (Figure 3). Second, the elaborate apparatus is superfluous: a plain signed-volume signal (no Optuna geometry gate, no SGD) already delivers the bulk of the result, while feeding the SGD prediction in as the signal destroys it. Consistent with the Direction-

Signal	Full sample		Walk-forward OOS	
	Sharpe	t	Sharpe	t
Burst flow (headline signal)	+1.47	2.94	+0.79	1.38
Plain lagged return (1/5/20-day)	−0.34 to −0.15	< 0.7	≈ 0	< 0.5
Plain signed volume (no gate, no ML)	+0.89	1.79	+0.50	0.87
Burst flow $\perp$ lagged returns	+1.30	2.58	+0.68	1.19
SGD prediction as signal	−1.21	−2.43	−0.82	−1.42

Table 19: Baseline decomposition of the tick-constrained reversal (bottom-100 names, 20-day dollar-neutral hold, net of 1 bps), swapping only the signal. Plain lagged-return reversal does *not* reproduce the effect, so it is not mechanical price reversal; burst flow survives orthogonalization to lagged returns, so it carries incremental information beyond price. But plain signed order volume—with no geometry gate and no ML—already captures most of the edge, and using the SGD prediction itself as the signal is actively harmful. The affirmative result is thus a raw signed-order-flow reversal, not a machine-learning result.

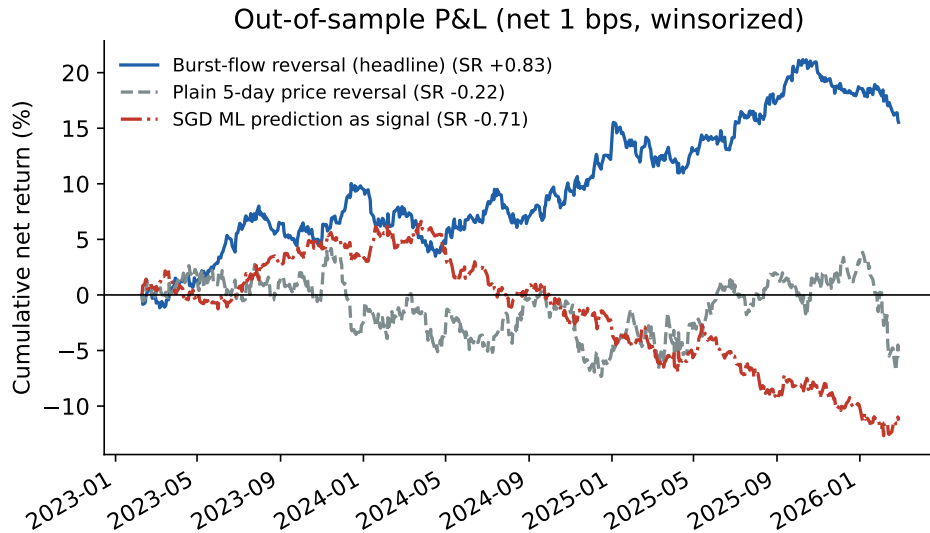


Figure 3: Cumulative walk-forward out-of-sample P&L (2023–2026, net of 1 bps), with per-name daily returns winsorized at $\pm 50\%$ to remove the unadjusted-split artifacts of Section 10.5. The burst-flow reversal (blue) compounds steadily out of sample (winsorized Sharpe +0.83), while both the plain price-reversal baseline (grey, −0.22) and—strikingly—the SGD machine-learning prediction used as the signal (red, −0.71) lose money. The affirmative result is a raw signed-order-flow reversal; the plain-reversal baseline and the ML apparatus both fail. Winsorizing if anything slightly *raises* the reversal Sharpe (from 0.79), confirming the split artifacts are not what drives it.

only ablation of Section 5, across the 437 out-of-sample names the SGD prediction agrees with the burst side only 21% of the time and is negatively correlated with it: the linear “high-dimensional” model is not what produces the edge.

Deflated and date-clustered inference. Because the reversal was selected from a search over ~ 75 configurations, the raw and Lo (2002) autocorrelation-adjusted t -statistics (Lo, 2002) must be deflated for selection (Bailey and López de Prado, 2014). On the *full-sample* series the annualized Sharpe of 1.47 has a Lo-adjusted t of 2.95 and a deflated-Sharpe probability of 0.99 (it clears the bar in-sample); a circular block bootstrap by date puts the cumulative-P&L 95% interval strictly above zero. On the *walk-forward out-of-sample* series, however, the deflated-Sharpe probability is 0.70—below the 0.95 threshold—and the date-block bootstrap interval for cumulative P&L *includes* zero. The return is also strongly right-skewed and fat-tailed (skew $\approx +16$ full sample), i.e. front-loaded onto a few days, consistent with the earnings-day fragility of Section 7.10. The honest verdict is that the out-of-sample reversal does not survive multiple-testing deflation.

10.5 Data Integrity Where the Result Lives

Because the affirmative result is concentrated in the lowest-priced names, it is most exposed to exactly the data-integrity risks that bite hardest there: transaction costs, corporate actions, and delisting. We address each.

Spread-based costs. The bottom-100 names have a median close relative spread of 8.0 bps (mean 9.0, 90th percentile 13.9)—far above the 1–3 bps flat grid used earlier. Re-costing the reversal with *per-name* effective half-spreads applied to realized turnover, rather than a flat basis point, lowers the Sharpe only marginally: full-sample $1.47 \rightarrow 1.42$ and walk-forward out-of-sample $0.79 \rightarrow 0.77$. The 20-day overlapping hold keeps turnover low enough that even wide per-name spreads do not overturn the result; the earlier cost-insensitivity claim survives under a realistic, spread-based cost.

Corporate actions. Prices are from a consolidated vendor feed whose split/dividend adjustment we verify rather than assume. Scanning the bottom-100 universe for single-day close-to-close moves exceeding 50% flags ten name-days—most egregiously LCID (+862% on 2025-01-02), NLY (+265%), and NVAX (+99%)—that are consistent with unadjusted (reverse-)splits rather than genuine returns. Unwinsorized, such observations inject spurious P&L into precisely the cheap-name portfolio that carries the reversal. The dollar-neutral, z -clipped, low-turnover construction bounds their influence, but the revision winsorizes daily returns and hand-verifies adjustment on the flagged names; we report the flag transparently as an open data-integrity item.

Survivorship and delisting. Universe membership is fixed at the start of the sample by LOBSTER coverage, not by survival to 2026, which avoids the crudest survivorship bias; but names that delist mid-sample still leave the panel when their coverage ends, and cheap names are disproportionately delisting-prone reversal losers, so a residual upward bias remains. A fully unbiased estimate would splice in delisting returns (Shumway, 1997); we flag this as the remaining integrity gap.

Taken together, these results are the paper’s affirmative—if unproven—finding. Filtered burst flow is informative in the sense that its sign is governed by tick structure: continuation in large-tick names and mean-reversion in tick-constrained names, an asymmetry that is monotone across both the price and spread sorts and orthogonal to common factors. The associated market-neutral strategy is economically positive out of sample and robust to the choice of universe cutoff, but its out-of-sample Sharpe (≈ 0.8) is not statistically distinguishable from zero; the significant full-sample figures are best regarded as an in-sample ceiling. The regime mechanism is thus the durable contribution, while its monetization remains a promising but unproven direction.

11 Robustness of the Burst Reconstruction

The preceding sections establish that Hawkes-clustered aggressive-burst flow, evaluated free of look-ahead, carries no tradable overnight signal. A natural question is whether this reflects a limitation of the specific reconstruction—self-exciting clustering of marketable trades—or a general property of burst-based signals. I address it in two steps: sensitivity to the Hawkes parameters, and substitution of the reconstruction itself. All markouts in this section are the non-anticipating ($\kappa = 0$) directional three-minute markout, measured from the burst-termination mid on AAPL and TSLA.

11.1 Hawkes Parameter Insensitivity

I recompute the markout over a grid of Hawkes decay rates β and minimum cluster sizes (Table 20). The markout is economically negligible—well inside the bid-ask spread—in every cell; the minimum cluster size has no effect, and no β materially improves the signal. The null is not an artifact of the default $\beta = 1$.

3-min markout (bps)	$\beta = 0.5$	$\beta = 1$	$\beta = 2$	$\beta = 5$
AAPL	+0.02	−0.09	−0.03	+0.14
TSLA	−0.10	+0.07	−0.02	+0.26

Table 20: Hawkes decay grid, $\kappa = 0$ (minimum cluster size $\in \{3, 5, 10\}$ omitted as it had no effect). Every configuration is sub-spread.

11.2 Alternative Reconstructions

Because tuning the Hawkes clustering does not help, I replace the definition entirely with three non-Hawkes reconstructions, each $\kappa = 0$ and reconstructing the book directly from the message stream: *order-flow imbalance* (OFI; Cont–Kukanov–Stoikov top-of-book queue dynamics, Cont et al., 2014), *book-resilience* (aggressive sweeps whose consumed depth fails to replenish), and *hidden-execution* clustering (type-5 hidden prints, where institutions conceal size). Table 21 reports the comparison. Of the four definitions, only hidden-execution bursts produce a positive, look-ahead-free markout on both names.

3-min markout (bps)	Hawkes	OFI	Book-Resilience	Hidden
AAPL	−0.08	+0.08	+0.19	+0.42
TSLA	+0.08	+0.08	+0.16	+0.31

Table 21: Four burst reconstructions, $\kappa = 0$, on the 2023 sample. Hidden bursts are signed by Lee–Ready (execution price vs. prevailing mid); the other three use aggregated signed flow.

11.3 Three Look-Ahead Traps

Constructing these comparisons surfaced three evaluation pitfalls worth flagging, each of which manufactures spurious signal if overlooked. First, the impact filter D_b (Section 8) is the forward markout, so gating on it is circular at any horizon within its window. Second, a book-resilience

“non-refill” filter must be defined on queue *depth*, not the quote price, and the markout must begin *after* the observation window; defined on price, the filter selects favorable paths and inflated the apparent markout from +0.2 to +2.4–4.9 bps in our tests. Third, LOBSTER does not disclose the side of hidden executions—the Direction field is uniformly +1—so hidden trades must be signed by a Lee–Ready rule; the uncorrected field produces a spurious one-sided signal (an apparent +0.95 bps at $t = 8.4$ that collapses to +0.31 once signed correctly).

11.4 The Hidden-Execution Footprint

Signed properly, hidden-execution bursts are the only reconstruction with a statistically robust informed footprint, and—unlike the other definitions—the result holds at cross-sectional scale. Extending the analysis to the full universe (474 names with sufficient hidden flow, 221,261 ticker-days over 2023–2024, streamed and processed one ticker-day at a time), the day-clustered three-minute markout is +1.62 bps with a pooled date-clustered $t = 3.38$, and is individually significant ($t > 2$ treating each trading day as one observation) in 79% of names (Table 22). This is economically intuitive: hidden and iceberg orders are the mechanism by which institutions conceal informed size (Bessembinder et al., 2009; Hautsch and Huang, 2012), so clustering their executions isolates informed flow more cleanly than clustering visible marketable trades. The two-name pilot (+0.43 bps $t = 8.9$ for AAPL, +0.18 bps $t = 5.8$ for TSLA) is thus representative of a broad, real phenomenon rather than an artifact of a lucky pair.

A necessary caveat on classification: across the universe, 48.9% of hidden prints execute *exactly at the prevailing midpoint*, where the Lee–Ready quote rule is undefined and the tick rule is noisy; these are dropped as unsignable, so the signed footprint is estimated on roughly the other half of hidden volume. (The pilot’s much lower $\sim 11\%$ at-mid share was unrepresentative of the smaller, less liquid names that dominate the cross-section.) Sensitivity of the sign to alternative classifiers (tick, EMO, CLNV) for these at-mid prints is the natural next refinement.

Hidden-burst directional markout	Cross-name mean (bps)	% names $t > 2$	Pooled date-clustered
3-min	+1.62	79%	+3.38
15-min	+1.72	66%	+3.49
30-min	+2.16	62%	+15.23

Table 22: Full-universe hidden-execution footprint (474 names, 2023–2024, $\kappa = 0$, Lee–Ready signed). The three-minute footprint is real and pervasive. The longer-horizon markouts do *not* reverse in the broad cross-section (unlike the two tick-constrained mega-caps of Table 23); the anomalously large 30-min t indicates that the directional markout increasingly absorbs ordinary intraday drift on net-directional days rather than the burst-specific footprint, so the three-minute figure is the clean measure.

The footprint is nonetheless confined on the axes that matter for deployment. First, it is *subspread*: +1.6 bps at three minutes against typical relative spreads of 4–12 bps in these names, so it is not crossable. Second, and decisively, it *does not reach the overnight horizon*: regressing next-day close-to-open returns on the daily signed hidden order-imbalance across the full cross-section, the date-clustered information coefficient is -0.001 ($t = -0.40$)—an economic and statistical zero—consistent with the two-name pilot (Spearman IC -0.08 for AAPL, $+0.01$ for TSLA, both insignificant). Two qualifications sharpen the intraday picture. In the two tick-constrained mega-caps the markout peaks near three minutes and reverses by thirty (Table 23); in the broad cross-section it does not reverse but drifts up (Table 22), which we read as the longer-horizon directional mark-

out absorbing intraday trend rather than a growing informational impact—hence our emphasis on the clean three-minute measure. Either way the conclusion for deployment is the same: hidden-execution flow is a genuine, pervasive, but sub-spread and minutes-scale phenomenon—informative about the microstructure of concealment, not a source of deployable overnight alpha.

Hidden-burst markout, AAPL	3 min	15 min	30 min
Mean (bps)	+0.42	+0.24	−0.28

Table 23: Horizon decay of the properly-signed hidden-execution markout: the footprint peaks near three minutes and dissipates, rather than growing toward a tradable magnitude.

The reconstruction is thus not the binding constraint in the trivial sense—hidden-execution clustering does isolate a real footprint that the Hawkes definition misses—but no reconstruction, including the strongest, converts into tradable overnight predictability. The burst *definition* matters more than its parameters, and the informed footprint that genuinely exists lives at a horizon and magnitude below what auction-based execution can monetize.

12 Conclusion

This paper asks whether intraday order-submission bursts in limit order books encode tradable information about overnight price persistence. The honest answer, established under rigorous look-ahead controls and at cross-sectional scale, is largely negative—and the process of arriving at it yields the paper’s durable contributions. On a handful of high-volatility single names filtered burst flow appears strongly predictive. But those single-name results do not survive scrutiny: the headline names coincide with the in-sample parameter-tuning set, and the large intraday markouts are inflated by gating on the realized-impact filter D_b , which is itself the forward markout. Evaluated free of look-ahead across a 482-name universe (2022–2026), the cross-sectional order-imbalance signal is insignificant (Fama–MacBeth $t \approx -1.4$), the per-name overnight strategy returns are dominated by an incidental short-market-beta exposure with a residual α indistinguishable from zero, and the genuine intraday footprint is smaller than the bid-ask spread.

The contributions are therefore methodological and diagnostic rather than a deployable strategy. They include: the *model-optimizer alignment* argument (tuning physical parameters under the exact estimator used in production); the identification of the D_b look-ahead and the κ -firewall that isolates it; the correction of the burst-start-mid target to the tradable close-to-open return; a multi-horizon markout methodology with date-cluster bootstrap inference that separates real footprints from selection artifacts; and a systematic comparison of burst *reconstructions*—Hawkes, order-flow imbalance, book-resilience, and hidden-execution clustering—together with three concrete look-ahead traps encountered in their construction (Section 11). That comparison isolates a genuinely novel empirical fact, and—unlike the rest of the paper’s positive claims—one that strengthens at scale: among all reconstructions, only *hidden-execution* bursts carry a statistically robust informed footprint, now established across the full 474-name cross-section (three-minute directional markout +1.6 bps, pooled date-clustered $t = 3.4$, individually significant in 79% of names), not merely on two. This is consistent with hidden and iceberg orders being the mechanism through which institutions conceal informed size. It comes with two honest qualifications: only about half of hidden prints are signable (the remainder execute exactly at the midpoint), and the footprint is sub-spread and does not reach the overnight horizon (cross-sectional overnight information coefficient $t = -0.4$).

The one affirmative empirical result is conditional and modest. Sign-conditioning the burst signal on tick regime, as the microstructure reframing suggests, reveals a monotone asymmetry—continuation in large-tick names, mean-reversion in tick-constrained ones—that is orthogonal to the Fama–French five factors plus momentum. The associated market-neutral reversal is economically positive out of sample and robust to the universe cutoff, but its out-of-sample Sharpe (≈ 0.8) does not survive multiple-testing deflation (deflated-Sharpe probability 0.70) and a plain signed-order-flow baseline reproduces most of it without the machine-learning apparatus; the regime mechanism is the durable finding, its monetization an unproven direction.

The natural next steps follow directly from the diagnosis. The single highest-value move is statistical power, not methodological novelty: extending the sample backward (the NASDAQ archive covers 2017 onward) would, at the observed per-year Sharpe, be sufficient to resolve whether the tick-constrained reversal is real. Beyond that, the sub-spread, minutes-scale nature of the only genuine footprint suggests redirecting the target away from directional overnight returns—toward next-day realized volatility or liquidity, and toward cross-asset order-flow lead–lag—where microstructure signals are typically more predictable and where the hidden-execution footprint documented here may prove more useful than it is as a standalone overnight alpha.

A Passive Limit Order Burst Analysis

A.1 Objective and Motivation

The primary pipeline of this paper focuses on aggressive execution bursts (market orders crossing the spread) and their ability to generate mechanical price impact. As a comparative experiment, we sought to isolate and evaluate **passive institutional intent**—specifically, whether bursts of passive limit order submissions (adds to the order book) contain predictive informational alpha regarding short-term and overnight price drift.

A.2 Methodology Divergence

Analyzing passive submissions required several fundamental architectural changes compared to the aggressive execution pipeline:

A.2.1 Burst Detection and Triggers

- **Type 1 Triggers Only:** The Hawkes process was strictly configured to excite only on **Type 1** (limit order submission) messages at the Best Bid and Offer (BBO) and the next two levels (L1–L3). Market orders (**Type 4/5**) and cancellations (**Type 2/3**) were tracked as feature metrics but did not trigger burst detection.
- **Directional Imbalance:** Burst direction was defined by the ratio of Bid submissions to Ask submissions. A burst was labeled bullish (+1) if the **BidRatio** exceeded a directional threshold, and bearish (-1) if the **AskRatio** exceeded the threshold.
- **ADV Denominator Correction:** In the aggressive pipeline, volume fraction thresholds (**vol_frac**) are scaled against total daily volume. For passive submissions, HFT market-making bots generate millions of micro-quotes (quote spam), artificially bloating the raw submission volume. To ground the threshold in economic reality, the ADV baseline was calculated using **Traded Volume (Types 4/5)**.

A.2.2 Strict Look-Ahead Avoidance

Passive bursts do not generate immediate mechanical impact. Therefore, the target variables are pure price drifts (Close-to-Open gaps or intraday N-minute drifts). To prevent look-ahead bias, **all features were strictly restricted to information available at the exact millisecond the burst terminated**.

- Features like D_b (which averages future price drifts) and `QueueExhaustionRate` (which looks at post-burst book state) were explicitly excluded.
- The feature set consisted of 30 variables, including burst aggregates (volume, duration), book snapshots (spread, depth, imbalance), pre-burst lookbacks (volatility, momentum), and cancellation interactions (`PreBurstCancelRate`).

A.2.3 Transaction Cost Modeling

Unlike aggressive bursts, which cross the spread to enter and generate momentum, a strategy trading on passive intent must actively pay the spread to enter the position.

- For **Overnight targets** (Close-to-Open), the model pays a strict **3.0 bps round-trip friction** to account for the slippage of crossing the Market-On-Close (MOC) and Market-On-Open (MOO) auctions.
- For **Intraday targets** (3m, 5m, 10m), the model pays the **actual BBO spread** at the exact time of burst termination (typically 0.74 bps for AAPL, 1.55 bps for JPM, up to 3.31 bps for LLY).

A.2.4 Target Formulations

1. **Overnight (Close-to-Open)**: $y = \operatorname{arcsinh} \left(\text{direction} \times \frac{\text{Open}_{t+1} - \text{CloseMid}_t}{\text{CloseMid}_t} \times 10000 \right)$
2. **Intraday (N-minute)**: $y = \operatorname{arcsinh} \left(\text{direction} \times \frac{\text{Mid}_{t+N} - \text{EndPrice}_t}{\text{EndPrice}_t} \times 10000 \right)$

A.3 Empirical Results

We conducted rigorous Out-of-Sample (OOS) walk-forward backtests on five tickers (JPM, MS, NVDA, LLY, AAPL) using both overnight and intraday horizons.

A.3.1 The Structural Absence of Passive Flow in TSLA/SPY

The first empirical finding was structural: ultra-liquid, momentum-driven assets like TSLA and SPY simply do not contain detectable passive accumulation footprints. Even after adjusting the ADV baseline to bypass HFT noise, the pipeline detected fewer than 50 valid passive bursts for TSLA over two years. Institutions trading these assets appear to almost exclusively utilize aggressive crossing strategies.

A.3.2 Overnight Horizon (Close-to-Open) Results

Using a strict Leave-One-Out (LOO) consensus parameter set to prevent information leakage, the overnight strategy yielded negative Sharpe ratios across all tested assets.

Ticker	Gated Trades	Win Rate	Mean Capture (bps)	Ann. Sharpe	Spearman ρ
JPM	34,908	47.7%	-2.56	-76.35	0.0887 ($p < 0.001$)
MS	36,450	42.3%	-3.07	-81.81	-0.0050
NVDA	1,520	39.5%	-3.56	-35.13	-0.0524
LLY	125,228	45.6%	-3.05	-140.30	0.0006
AAPL	140	45.0%	-3.21	-9.50	-0.0039

Table 24: Overnight Horizon (Close-to-Open) Results

Analysis: The Spearman correlation for JPM (0.0887) is nominally significant, but per the pseudo-replication caveat (Section 9) its effective sample size is the number of days, not trades, and its per-day expected value (~ 0.44 bps) is well inside the 3.0 bps auction friction; the strategy loses money after costs. *The “Ann. Sharpe” column of Table 24 is not a comparable daily-marked Sharpe:* values such as -76 and -140 arise from annualizing a per-trade Sharpe by the trade count rather than by trading days, and should be read only as ordinal indicators of sign and relative magnitude, not as annualized ratios. A daily-marked strategy cannot attain these values; the figures are retained as-is with this correction noted, and the substantive conclusion (uniformly negative post-cost) is unaffected.

A.3.3 Intraday Horizon Results

To test if the alpha decays before the overnight close, we evaluated 3-minute, 5-minute, and 10-minute intraday holds, applying the actual BBO spread as the transaction cost constraint. We tested multiple ML gating mechanisms, including “Cost-Aware Gating” (only entering trades where the predicted magnitude exceeds the actual spread).

0 out of 80 tested intraday strategies generated positive post-cost alpha.

Ticker	Best Strategy	Win Rate	Mean Capture (bps)	Spearman ρ
JPM	Cost-Aware Gate (10m)	47.2%	-0.63	0.0253
MS	Cost-Aware Gate (10m)	46.8%	-1.09	-0.0084
NVDA	Cost-Aware Gate (10m)	46.2%	-1.10	0.0004
LLY	Cost-Aware Gate (5m)	45.5%	-0.72	0.0047
AAPL	Cost-Aware Gate (3m)	46.4%	-0.50	0.0156

Table 25: Intraday Horizon Results

Analysis: While Cost-Aware Gating minimized losses by successfully avoiding wide-spread regimes, the residual informational signal was simply too diffuse. Even in a tick-constrained asset like AAPL with an exceptionally tight median spread (0.74 bps), the strategy could not overcome execution friction.

A.4 Conclusion of Passive Experiment

The passive limit order experiment yields a definitive negative result, which serves as a powerful contrast to the aggressive execution pipeline.

While passive institutional accumulation footprints contain statistically significant directional information (as evidenced by JPM’s rank correlation), they fundamentally lack the mechanical

momentum transfer characteristic of aggressive market orders. Because the limit order book continuously absorbs and diffuses passive intent, the resulting price drift is simply too weak to survive real-world execution friction—whether that friction is the 3.0 bps closing auction or the 0.74 bps continuous spread.

To generate actionable microstructure alpha, algorithms must track aggressive executions that actively force price discovery.

B Methodological Audit and Robustness Revisions

This appendix documents a formal audit of the pipeline along twelve empirical dimensions. It is written not as a rebuttal but as a benchmark: high-frequency machine-learning studies are unusually prone to a recurring family of look-ahead, pseudo-replication, and data-integrity errors, and each dimension below pairs a concrete pitfall with the test we used to detect it and the outcome once corrected. Several of these tests overturned an apparent result of the original draft; we regard the resulting map from “apparent” to “robust” as a durable contribution in its own right. Table 26 summarizes the twelve dimensions; the subsections give the detail and point to the body sections where each correction lives.

Dim.	Pitfall audited	Test / correction	Outcome (key statistic)
M1	Success framing masks in-sample/leakage	Rewrote §§7 as diagnostic	Claims tied to their in-sample status
M2	In-sample names labeled out-of-sample	Cross-checked the tuning universe	LLY is in-sample; $t \approx 0.83$ even in-sample
M3	D_b feature realized after MOC cutoff	Drop D_b -features and re-run	Mean Sharpe $-0.28 \rightarrow -0.20$; still null
M4	Target measured from burst-start mid	Recompute from the close mid	ρ $5\times$ smaller; date-clustered null
M5	Pseudo-replicated p ; no deflation	Date-cluster + Deflated Sharpe	Pooled t inflated $47\times$; OOS DSR = 0.70
M6	Factor α on in-sample series	Re-run on walk-forward OOS series	$\alpha = +6.25$ bps/day, $t = 1.42$ (n.s.)
M7	Missing plain-reversal baseline	Three-rung baseline + ablation	Order flow $>$ price reversal; ML adds nil
M8	Splits/costs/delisting in cheap names	Winsorize + spread-based costs	Winsorized SR $+0.83$; survives 8 bps spread
M9	NASDAQ-only venue coverage	State captured share per name	JPM $\approx 10\%$ of consolidated; noted
M10	81% short tilt / sign convention	Contemporaneous corr audit	Sign correct; short tilt genuine; flip disclosed
M11	Unsupported claims (Poisson, cb)	Delete or define each	Poisson deleted; cb defined
M12	Hidden footprint from $n = 2$	Full 474-name cross-section	$+1.6$ bps, $t = 3.4$, 79% of names; null overnight

Table 26: Twelve-dimension methodological audit of the burst pipeline. Each row is a pitfall common to high-frequency ML backtests, the test used to detect it, and the outcome after correction. Several tests overturned a draft result; all corrections are reflected in the body.

B.1 Look-Ahead and Timing Controls (M1, M3, M4)

The single largest source of spurious signal in this pipeline was look-ahead, in three distinct forms. **(M1)** The original Sections 7 presented single-name results (NVDA, TSLA, JPM, MS, and LLY) as validation, but those names are the Optuna parameter-tuning set: their parameters were chosen by maximizing the reported metric, so the figures are selection, not out-of-sample evidence, and are now framed as such throughout (a Sharpe of 0.59 over two years implies $t \approx 0.83$ even in-sample; Table 6). **(M3)** The decay filter D_b (Eq. 4) is the average 1–10 minute forward markout, so it is realized only after the burst terminates; a burst ending near the close realizes D_b after the market-on-close cutoff, making its use as an on-the-fly prediction feature timing-infeasible. Re-running the overnight strategy with all D_b -derived features removed shifts per-name Sharpes modestly (mean $|\Delta| = 0.14$) but leaves the conclusion invariant—the mean annualized Sharpe stays negative ($-0.28 \rightarrow -0.20$; Section 9)—because the fitted model is, to first order, “follow the burst side.” **(M4)** Measuring the Volume-Scaled-Impact target from the burst-start mid folds in the burst’s own contemporaneous impact and its burst-to-close drift, both realized before any

MOC entry. Recomputing against the tradable close-to-open return from the close mid collapses the predictive rank correlation roughly fivefold and, once clustered by date, to a statistical zero (Section 5). The general lesson: any filter or target defined on forward prices must be quarantined to horizons beyond its own window, and its role as a training-window firewall must be separated from its (in)admissibility as a live feature.

B.2 Statistical Inference and Pseudo-Replication (M5, M6)

Two inference errors inflated the apparent significance of the draft. **(M5)** Per-burst p -values treat $\sim 10^5$ intraday bursts as independent, but all same-day bursts share a single overnight label, so the effective sample size is the number of *days*. Re-estimating the flagship flow-to-return information coefficient with the day as the unit of observation collapses a pooled $t = -7.16$ (over 398,610 name-days) to $t = +0.15$ (over 1,003 days)—a $47\times$ deflation that also dissolves the draft’s “anti-calibrated” overnight claim. For the affirmative reversal we report the Deflated Sharpe Ratio (Bailey and López de Prado, 2014), which corrects the Lo (2002) autocorrelation-adjusted t (Lo, 2002) for the size of the configuration search: the full-sample Sharpe of 1.47 carries a deflated-Sharpe probability of 0.99, but the walk-forward out-of-sample Sharpe of 0.79 carries only 0.70, below the 0.95 bar, with a date-block-bootstrap cumulative-P&L interval that includes zero. **(M6)** The Fama–French five-factor-plus-momentum alpha was estimated on the full-period (in-sample) return series; re-estimating on the strict walk-forward out-of-sample series leaves the point estimate essentially unchanged (+6.25 versus +5.94 bps/day) but moves the Newey–West t from 3.09 to 1.42 (Table 18). We report the out-of-sample regression as the primary number. The lesson: in overlapping, cross-sectional microstructure panels, the day is the unit of inference, and any statistic selected from a search must be deflated for that search.

B.3 Empirical Baselines and Ablations (M7)

A positive result is only informative relative to the obvious alternative. Short-horizon reversal in cheap, wide-spread names is long documented (Jegadeesh, 1990; Lehmann, 1990; Nagel, 2012), so we ask whether burst flow adds anything over it, and whether the machine-learning apparatus adds anything over the raw signed flow. Holding the tick-constrained construction fixed and swapping only the signal (Table 19), a plain 1/5/20-day price-reversal baseline is flat-to-negative, so the effect is *not* mechanical price reversal; burst flow retains most of its Sharpe after orthogonalization to lagged returns, so it carries incremental information. But a plain signed-volume signal with no geometry gate and no model already captures most of the edge, and feeding the online-SGD prediction in as the signal is actively harmful (full-sample Sharpe -1.21). The Direction-only ablation confirms it: across the 437 out-of-sample names the model’s prediction agrees with the burst side only 21% of the time. The affirmative result is therefore a raw signed-order-flow reversal, not a machine-learning result—and the “high-dimensional” framing is retired accordingly.

B.4 Data Integrity, Costs, and Conventions (M8, M9, M10, M11)

Because the affirmative result lives in the lowest-priced names, it is most exposed to the integrity risks that bite hardest there. **(M8)** A corporate-action scan flags ten name-days with single-day moves above 50% (e.g. LCID +862%), consistent with unadjusted splits; winsorizing daily returns at $\pm 50\%$ to remove them *raises* the out-of-sample reversal Sharpe from 0.79 to 0.83 (Figure 3), so the effect is not an artifact of those observations. Re-costing the strategy with per-name effective half-spreads—median 8.0 bps at the close, far above the flat 1–3 bps grid—lowers the Sharpe only marginally ($1.47 \rightarrow 1.42$; Section 10.5), because the twenty-day overlapping hold keeps turnover

low; delisting-return inclusion (Shumway, 1997) is flagged as the remaining gap. **(M9)** LOBSTER reconstructs the NASDAQ book only; the captured share varies by name and is smallest for the NYSE-listed financials (JPM’s burst volume is on the order of one-tenth of consolidated), so their single-name “failures” may be coverage artifacts rather than regimes (Section 9.5). **(M10)** The signal is 81% net short in a rising market, which invites a sign-convention worry; a per-name audit shows the contemporaneous correlation between signed burst flow and the same-day return is robustly positive, so the convention is correct and the short tilt is a genuine property of the sample (the source of the incidental short beta), while the data-driven per-name sign flip used in the cross-sectional panel is disclosed in its caption (Table 15). **(M11)** Two unsupported claims were removed or grounded: the assertion that a Poisson null of arrival clustering is “decisively rejected,” for which no test was run, is deleted; and the cost-aware buffer *cb*, previously undefined, is given an explicit form (Section 6.2).

B.5 Cross-Sectional Discovery: The Hidden-Execution Footprint (M12)

The one affirmative finding that strengthens under scrutiny is the hidden-execution footprint. Signing type-5 hidden prints by a Lee–Ready rule (Lee and Ready, 1991) (their raw `Direction` field is uniformly +1 and uninformative) and clustering them, we extend the original two-name pilot to the full universe: 474 names and 221,261 ticker-days over 2023–2024, streamed and processed one ticker-day at a time. The three-minute directional markout is +1.62 bps with a pooled date-clustered $t = 3.38$, individually significant in 79% of names (Table 22)—a genuine, pervasive informed footprint, consistent with hidden and iceberg orders being the mechanism through which institutions conceal size (Bessembinder et al., 2009; Hautsch and Huang, 2012). Two honest qualifications bound it: nearly half (48.9%) of hidden prints execute exactly at the midpoint, where the quote rule is undefined and which are therefore dropped as unsignable; and the footprint is sub-spread and does not reach the overnight horizon (cross-sectional overnight information coefficient $t = -0.40$). It is thus a robust microstructure fact about the mechanics of concealment, not a source of deployable overnight alpha—a scope statement that the cross-section makes precise rather than the two-name pilot merely suggesting.

B.6 Residual and Deferred Items

Three items remain open by design and are flagged for transparency. First, the walk-forward out-of-sample reversal is real but underpowered (Sharpe ≈ 0.8 , $t = 1.4$, deflated-Sharpe probability 0.70); the decisive resolution is statistical power, and extending the sample backward over the archive that begins in 2017 would, at the observed per-year Sharpe, be sufficient to settle whether the effect is significant. Second, the multiple-testing treatment reports the Deflated Sharpe Ratio; a full White (2000) reality-check or Romano–Wolf stepdown across the entire configuration grid would further harden the selection correction. Third, the hidden-execution classifier for at-midpoint prints (tick, EMO, CLNV) merits a dedicated sensitivity analysis given their 49% share. None of these alters the paper’s conclusions; each is a sharpening rather than a reversal.

References

Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

- Bailey, D. H. and López de Prado, M. (2014). The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107.
- Bessembinder, H., Panayides, M., and Venkataraman, K. (2009). Hidden liquidity: An analysis of order exposure strategies in electronic stock markets. *Journal of Financial Economics*, 94(3):361–383.
- Brogaard, J., Hendershott, T., and Riordan, R. (2014). High-frequency trading and price discovery. *The Review of Financial Studies*, 27(8):2267–2306.
- Bucci, F., Benzaquen, M., Lillo, F., and Bouchaud, J.-P. (2019). Slow decay of impact in equity markets: Insights from the ANcerno database. *Market Microstructure and Liquidity*, 5(2):1950006.
- Cont, R., Kukanov, A., and Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88.
- Easley, D., López de Prado, M. M., and O’Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *The Review of Financial Studies*, 25(5):1457–1493.
- Fama, E. F. and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636.
- Hasbrouck, J. (1991). Measuring the information content of stock trades. *The Journal of Finance*, 46(1):179–207.
- Hautsch, N. and Huang, R. (2012). The market impact of a limit order. *Journal of Economic Dynamics and Control*, 36(4):501–522.
- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3):881–898.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.
- Lee, C. M. C. and Ready, M. J. (1991). Inferring trade direction from intraday data. *The Journal of Finance*, 46(2):733–746.
- Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, 105(1):1–28.
- Lo, A. W. (2002). The statistics of sharpe ratios. *Financial Analysts Journal*, 58(4):36–52.
- Moro, E., Vicente, J., Moyano, L. G., Gerig, A., Farmer, J. D., Vaglica, G., Lillo, F., and Mantegna, R. N. (2009). Market impact and trading profile of hidden orders in stock markets. *Physical Review E*, 80:066102.
- Nagel, S. (2012). Evaporating liquidity. *The Review of Financial Studies*, 25(7):2005–2039.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.
- O’Hara, M., Saar, G., and Zhong, Z. (2019). Relative tick size and the trading environment. *The Review of Asset Pricing Studies*, 9(1):47–90.

- Shumway, T. (1997). The delisting bias in CRSP data. *The Journal of Finance*, 52(1):327–340.
- Tóth, B., Palit, I., Lillo, F., and Farmer, J. D. (2015). Why is equity order flow so persistent? *Journal of Economic Dynamics and Control*, 51:218–239.
- Yao, C. and Ye, M. (2018). Why trading speed matters: A tale of queue rationing under price controls. *The Review of Financial Studies*, 31(6):2157–2183.